

# Clusterwise Linear Regression with an Unknown Fraction of Outliers:

## Exact Subsample Fits, Sequential Flagging and Group Recovery

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### Abstract

Trimming methods for robust clusterwise linear regression discard a fixed fraction of the data, and their answer depends on that fraction, which is rarely known. We propose a procedure that estimates it instead. The procedure fits the model exactly, by branch and bound, on many small random subsamples. Units far from the best fit, in the response or in the covariates, are flagged, and later subsamples are drawn from the unflagged units only. A second step asks whether the flagged units are contamination or a group that the fit missed: it looks for a line among them and compares the two descriptions by the likelihood of Gaussian groups plus uniform noise. We give exact results for one stage of the subsampling, the large-sample limit of the final reweighting step, and the population value of the second step. In a simulation study whose criteria were fixed before the data were analysed, the procedure was never more than 0.02 below the most accurate of eight competitors in seven designs with up to 20% of outliers, and every fixed trimming level failed in some design at up to 35%. The second step recovered groups of 15% and 20% of the data that the first step took for contamination. The procedure breaks down from 25% to 30% of outliers with uniform noise, high-leverage points or three groups; with two groups at high contamination, mixtures with contaminated normal or Laplace errors are more robust.

**Keywords:** clusterwise regression; robust clustering; trimming; outliers; subsampling; branch and bound; mixture of regressions.

## 1 Introduction

In clusterwise linear regression each observation follows one of  $K$  linear models, and the model that generated it is not observed. For units  $(x_i, y_i)$ ,  $i = 1, \dots, n$ , with  $x_i \in \mathbb{R}^d$  including an intercept,

$$y_i = x_i^\top \theta_{z_i}^0 + \sigma_{z_i} e_i, \quad z_i \in \{1, \dots, K\}, \quad (1)$$

where  $z_i$  is the unobserved label of unit  $i$ ,  $\theta_k^0 \in \mathbb{R}^d$  and  $\sigma_k > 0$  are the coefficients and noise scale of group  $k$ , and the  $e_i$  are independent standard errors. The model goes back to Quandt (1972) and Späth (1979). It is usually fitted by least squares over partitions or by maximum likelihood for a mixture of regressions (DeSarbo & Cron, 1988; De Veaux, 1989).

Both fits break down when a fraction of the data does not follow any of the  $K$  models. A few gross outliers in the response can pull a fitted line far from every group, or be fitted as a group of their own. The standard remedy is trimming. TCLUS-REG (García-Escudero et al., 2010) discards the fraction  $\alpha$  of units that fit worst and fits the model to the rest; the trimmed likelihood

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estimator (Neykov et al., 2007) and the trimmed cluster-weighted model (García-Escudero et al., 2017) do the same within other likelihoods. Each needs the trimming level  $\alpha$ . If  $\alpha$  is smaller than the fraction of outliers, the fit breaks down. If  $\alpha$  is much larger, genuine units are discarded, and a small group can be trimmed away altogether. The analyst rarely knows the right value.

Several ways around this choice have been proposed. One trims at a deliberately high level and then reinstates the units that are well fitted, a reweighting step proposed for robust clustering by Dotto et al. (2018). Another monitors the fit over a grid of levels and lets the analyst choose (Torti et al., 2019). A third adapts the trimming within each group (Chang et al., 2020), and a fourth replaces trimming by bounded loss functions (Bai et al., 2012). The first needs a level that is high enough and the second an analyst’s judgement; the evidence below shows that the third and fourth break down with three groups even without outliers, and at moderate contamination in other designs. Mixtures with a mean-shift parameter for each unit (Yu et al., 2017), with a uniform noise component (Banfield & Raftery, 1993), or with contaminated normal errors whose proportion of bad points is estimated (Punzo & McNicholas, 2017; Mazza & Punzo, 2020) model the outliers instead of trimming them, and so need no level either. Outside statistics, multiple-model fitting from random minimal samples, from RANSAC (Fischler & Bolles, 1981) to J-linkage (Toldo & Fusiello, 2008), addresses a related problem in computer vision, with thresholds set by the user.

This paper proposes a procedure that estimates the fraction of outliers instead of asking for it. It rests on one observation. If the model is fitted *exactly*, by least squares over all partitions, to a small random subsample that happens to contain no outlier, the fit is a clean fit of the model, and it identifies the outliers of the whole sample as the units far from every fitted line. The procedure therefore draws many small subsamples, fits each one exactly, scores each fit by a bounded loss on the units not yet flagged, and flags the units that lie far from the best fit. Later subsamples are drawn only from units not yet flagged, so that clean subsamples become more frequent as the procedure runs. The fit is refitted on the unflagged units and ends with the reweighting step used in robust regression (Rousseeuw & Leroy, 1987). Units whose covariates are outlying are flagged by a separate screen. A unit is flagged when its residual exceeds a fixed multiple of a robust scale, as in reweighted least trimmed squares; this cut-off takes the place of the trimming level, and it does not depend on how many outliers there are.

A procedure that decides for itself what to flag must also decide what a group is. When one group is small, the best fit may spend two lines on the large group and flag the small one as contamination; our simulations show this happening to a group of 20% of the data even without outliers. The second step of the procedure therefore looks at what the first step flagged. It searches the flagged units for a line around which many of them are concentrated, and it keeps the line if the likelihood of the model behind the flagging rule, Gaussian groups plus uniform noise, prefers the fit that contains it. A trimming method cannot ask this question, because what it trims is fixed in advance by the level.

Drawing subsamples to find a clean one is the idea behind least median of squares and least trimmed squares (Rousseeuw, 1984; Rousseeuw & Van Driessen, 2006), and TCLUS-REG starts its own search from random subsets. What is different here is fourfold. Each subsample contains several units of every group and is solved to global optimality, so a clean subsample yields the least-squares clusterwise fit of that subsample and not a fit to  $d$  points. The subsampling is sequential and removes the flagged units from later draws, which raises the chance of a clean subsample from one stage to the next. The trimmed fraction is an output: the fraction of flagged units estimates the contamination. And the flagged set is examined for a missed group. The operations research literature treats clusterwise regression as an optimisation problem: mixed-integer formulations (Bertsimas & Shioda, 2007), exact branch and bound (Carbonneau et al., 2011, 2012), probabilistic branch and bound (Fois et al., 2026), nonsmooth optimisation (Bagirov et al., 2013; Joki et al.,

2020) and heuristics for large samples (Park et al., 2017). These methods fit the whole sample and are evaluated on data without gross outliers; the exact ones handle a few hundred units. Here the exact fit is applied to subsamples of a dozen units, where it is fast, and the whole sample is used only to score and flag. The exact subsample fit is thus used as a building block of a robust estimator, and our ablation shows when exactness matters.

The contributions are the following.

- (i) A two-step procedure for clusterwise linear regression that is robust to outliers in the response and in the covariates, estimates their fraction instead of requiring it, and checks whether what it flagged is a group (Section 2). Its subsamples are solved exactly by branch and bound; an ablation shows that heuristic subsample fits lose accuracy with three or four groups or three covariates, while the exact fit takes milliseconds (Section 4.4 and Appendix B).
- (ii) Three partial results (Section 3). Conditionally on the data, the subsamples of one stage are independent, and the probability that the stage contains a clean subsample is given exactly in terms of the contamination among the units still eligible. The final reweighting step, applied at the true parameter, converges to a limit computed from the clean units, within a distance of the order of the overlap between groups, and the fraction it flags converges to the contamination plus an explicit term for clean units. The likelihood comparison of the second step gains, per unit of a missed group, the log-ratio of the group’s density to the noise density, and loses almost nothing when it merges two lines that share one group.
- (iii) A simulation study on 6,700 data sets with three sets of criteria fixed before the data they concern were analysed, against five robust competitors run from their authors’ software and two mixtures that need no level, one with Laplace errors and one with a uniform noise component (Section 4). In each of the seven designs without high-leverage points and with up to 20% of outliers, the procedure is never more than 0.02 below the best competitor chosen afterwards, and no competitor at a fixed trimming level, with or without reweighting, is as good in every design. The second step recovers groups of 15% and 20% of the data that the first step alone takes for contamination.

The procedure is not a better way to fit clean data. Without outliers, least squares with random restarts is as accurate and faster, and the procedure is useful only when outliers may be present. It also has limits, which Section 4 reports: it breaks down from 25% of outliers when these are spread uniformly over the data range or lie at high leverage, and from 30% when there are three groups. With two groups at high contamination, uniform noise included, a mixture of regressions with Laplace errors (Song et al., 2014) is more robust, and with three groups at 35% of outliers TCLUST-REG at a high level is. Its tuning constants are multiples of a scale and conventional thresholds, not fractions of the data, and we used the same values in every experiment.

## 2 The procedure

### 2.1 Exact fits on small subsamples

For a set  $S$  of units, the least-squares clusterwise fit of  $S$  solves

$$\min_{z \in \{1, \dots, K\}^S} \min_{\theta \in \mathbb{R}^{Kd}} \sum_{i \in S} (y_i - x_i^\top \theta_{z_i})^2. \quad (2)$$

For fixed labels the inner problem separates into  $K$  ordinary least-squares fits, so the difficulty lies in the labels alone. On the whole sample the problem is hard: with  $x_i \equiv 1$  it is  $k$ -means on the

line, and  $k$ -means is NP-hard in general (Aloise et al., 2009; Mahajan et al., 2012). On a subsample of a few dozen units it can be solved exactly. We use a depth-first branch and bound over label vectors, which prunes a partial labelling as soon as the sum of the residual sums of squares of its groups exceeds the best value found (Appendix B). Table 1 gives its cost. The pruning removes most of the search: with four groups and  $m = 16$  the median search visits about  $3 \times 10^4$  nodes where plain enumeration would list  $2 \times 10^8$  labellings, and takes about 7 milliseconds. The cost does not depend on  $n$ , but it grows quickly with  $K$  and  $m$ , and this limits the procedure to a small number of groups.

Table 1: Cost of the exact fit of one subsample by branch and bound: nodes visited and time in milliseconds (median and 90th percentile over 200 subsamples of ten simulated data sets with 10% of outliers), and the number  $K^m/K!$  of labellings that plain enumeration would list.

| Design     | $K$ | $d$ | $m$ | Nodes, median     | Nodes, 90%        | Time (ms), median | $K^m/K!$             |
|------------|-----|-----|-----|-------------------|-------------------|-------------------|----------------------|
| D1         | 2   | 2   | 8   | 91                | 133               | 0.03              | 128                  |
| D1         | 2   | 2   | 12  | 309               | 634               | 0.08              | $2.0 \times 10^3$    |
| D1         | 2   | 2   | 16  | $1.0 \times 10^3$ | $6.9 \times 10^3$ | 0.22              | $3.3 \times 10^4$    |
| D2         | 3   | 2   | 12  | $1.7 \times 10^3$ | $2.9 \times 10^3$ | 0.43              | $8.9 \times 10^4$    |
| D2         | 3   | 2   | 16  | $8.1 \times 10^3$ | $2.2 \times 10^4$ | 1.71              | $7.2 \times 10^6$    |
| four lines | 4   | 2   | 16  | $3.0 \times 10^4$ | $6.4 \times 10^4$ | 7.25              | $1.8 \times 10^8$    |
| four lines | 4   | 2   | 20  | $1.6 \times 10^5$ | $6.7 \times 10^5$ | 31.50             | $4.6 \times 10^{10}$ |
| D7         | 2   | 4   | 12  | 606               | 815               | 0.57              | $2.0 \times 10^3$    |

An exact solution has a simple structure. Write  $h_\theta(u) = \arg \min_k (y - x^\top \theta_k)^2$  for the index of the surface nearest to a unit  $u = (x, y)$ , ties being broken at random.

**Proposition 1.** *A pair  $(\hat{z}^S, \hat{\theta}^S)$  solves (2) if and only if  $\hat{\theta}^S$  minimises  $\sum_{i \in S} \min_k (y_i - x_i^\top \theta_k)^2$  and every unit of  $S$  is assigned to its nearest surface,  $\hat{z}_i^S = h_{\hat{\theta}^S}(u_i)$ , up to ties.*

The proof is one line: for fixed  $\theta$  the best label of each unit is its nearest surface. The exact fit of a subsample is thus a  $K$ -means fit with surfaces in place of centres (Pollard, 1981). Assigning each of the  $n$  units to its nearest fitted surface, by the rule  $h_{\hat{\theta}^S}$ , extends the fit to the whole sample. We call the fit of one subsample a *replicate*.

A replicate is *admissible* if each of its  $K$  groups contains at least  $d + 1$  units. Inadmissible replicates are discarded and redrawn. When the groups are unbalanced, admissibility forces  $m\pi_{\min} \gtrsim d + 1$ , where  $m$  is the subsample size and  $\pi_{\min}$  the smallest group proportion.

## 2.2 Sequential flagging

Algorithm 1 states the first step of the procedure. It runs in  $L$  stages. Each stage draws  $B/L$  admissible replicates, each on a subsample of size  $m$  drawn uniformly from the units not yet flagged. After each stage, all replicates drawn so far are scored by a bounded loss on the units not yet flagged, the best-scoring replicate becomes the reference, and the units far from the reference are flagged. The final fit is computed from the unflagged units.

*Flagging.* A unit is far from a fit  $\theta$  in the response if its absolute residual from its nearest surface exceeds  $c$  times the scale of the group of that surface. The scale of group  $k$  is 1.4826 times the median absolute residual of the units assigned to surface  $k$  (or of all units, when fewer than ten are assigned to  $k$ ), so a group with small noise does not hide outliers and a group with large noise is not trimmed. A unit is far in the covariates if its robust distance from the other units of its group, computed with the minimum covariance determinant (Rousseeuw & Van Driessen, 1999)

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**Algorithm 1** Clusterwise regression from exactly solved subsamples, with sequential flagging

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**Require:** data  $(x_i, y_i)_{i \leq n}$ ; number of groups  $K$ ; subsample size  $m$ ; replicates  $B$ ; stages  $L$ ; weight  $\lambda$ ; cut-offs  $c, c_b$

- 1:  $\mathcal{F} \leftarrow$  units whose covariates are outlying (robust distance above the 0.975 quantile of  $\chi_p^2$ ,  $p$  the number of covariates)
- 2: **for**  $\ell = 1, \dots, L$  **do**
- 3:   draw  $B/L$  admissible replicates, on subsamples of size  $m$  drawn uniformly from  $\{1, \dots, n\} \setminus \mathcal{F}$ , each solved exactly
- 4:   for every replicate  $b$  so far and every unit  $i$ :  $r_{bi} \leftarrow \min_k (y_i - x_i^\top \hat{\theta}_k^{(b)})^2$
- 5:    $s^2 \leftarrow \min_b \text{med}_{i \notin \mathcal{F}} r_{bi} / 0.4549$   $\triangleright 0.4549 = \text{med } \chi_1^2$
- 6:    $F_b \leftarrow \sum_{i \notin \mathcal{F}} \min\{r_{bi}, c_b^2 s^2\}$ ;  $w_b \leftarrow \exp\{-\lambda(F_b - \min_{b'} F_{b'}) / (|\mathcal{F}^c| s^2)\}$
- 7:   reference  $\leftarrow$  replicate  $b^* = \arg \min_b F_b$
- 8:    $\mathcal{F} \leftarrow \{i : i \text{ is far from the reference in the response or in the covariates}\}$  (*Flagging* below)
- 9: **end for**
- 10: labels  $\tilde{z} \leftarrow$  plurality of the nearest-surface labels of all replicates, weighted by  $w$
- 11: refit each group on its unflagged units; concentration steps keeping  $n - |\mathcal{F}|$  units
- 12: reweighting: recompute the flags at the current fit and refit on the unflagged units
- 13: **return** coefficients  $\hat{\theta}$ , labels  $h_{\hat{\theta}}(u_i)$  for every unit, the flags  $\mathcal{F}$  recomputed at  $\hat{\theta}$ , and  $\hat{\alpha} = |\mathcal{F}|/n$

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(the median and the median absolute deviation when there is one covariate), exceeds the 0.975 quantile of  $\chi_p^2$ , with  $p$  the number of covariates; the test is applied only to groups with at least  $\max(10, 5p)$  units. This second test is the covariate trimming of TCLUST-REG (García-Escudero et al., 2010), with a fixed quantile in place of a trimming level. At the start, before any fit, the covariate test is applied to all units as one group.

*Scoring.* The bounded loss caps each squared residual at  $c_b^2 s^2$ , where  $s$  is a scale common to all replicates. A replicate that fits the clean units well has a low score whatever the outliers do, because each outlier contributes at most the cap. The scale is the smallest, over the replicates, of the median squared residual divided by 0.4549, the median of  $\chi_1^2$ : for a replicate that fits the clean majority, this is a rough estimate of the error variance, and taking the smallest value makes the scale that of the best replicate rather than of a typical one. A replicate that overfits a subsample does not lower it much, because the median is taken over all unflagged units, not over the subsample.

*Labels and final fit.* The labels of the final fit are the weighted plurality of the labels that the replicates give to each unit. Replicates are aligned before voting by solving an assignment problem between their coefficients and those of a common reference, solved by enumerating the  $K!$  relabellings (or by the Hungarian method (Kuhn, 1955) for larger  $K$ ). The reference is a pilot replicate, the best of ten preliminary replicates by the median of the squared residuals. It serves only for alignment; it does not vote. After the vote, a group with fewer than  $d$  unflagged units keeps the coefficients of the best-scoring replicate  $b^*$  of the last stage. The concentration steps are those of least trimmed squares (Rousseeuw & Van Driessen, 2006) and TCLUST-REG, with the number of retained units set by the flags rather than by a level. The last step is the reweighting step of robust regression (Rousseeuw & Leroy, 1987; Gervini & Yohai, 2002): the flags are recomputed at the current fit, and each group is refitted on its unflagged units. Every unit, flagged or not, then receives the label of its nearest fitted surface.

### 2.3 Is the flagged set a group?

Algorithm 1 flags every unit that its  $K$  lines do not explain, and it cannot tell a unit that belongs to no group from a unit of a group it has missed. The second case arises when one group is small. The best-scoring fit may then spend two nearly parallel lines on the large group, whose units it fits more tightly that way, and flag most of the small group (Section 4.5). Algorithm 2 examines the flagged set for this case after Algorithm 1 has finished.

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**Algorithm 2** Group recovery, applied to the fit of Algorithm 1

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**Require:** fit  $\hat{\theta}$  with flags  $\mathcal{F}$  and group scales  $s_1, \dots, s_K$ ; noise range  $R = 1.1 (\max_i y_i - \min_i y_i)$ ; margin  $\frac{d+1}{2} \log n$ ; restriction factor 12

- 1: stop if  $|\mathcal{F}| > n/3$ ; otherwise let  $\mathcal{F}'$  be  $\mathcal{F}$  without the units of the initial covariate screen, and  $s = \text{med}_k s_k$
- 2: fit a line  $\beta$  to  $\mathcal{F}'$ : the line through  $d$  units of  $\mathcal{F}'$ , among 500 random choices, that has most units of  $\mathcal{F}'$  within  $cs$ , refitted three times by least squares on the units within  $cs$
- 3: stop unless at least  $\max\{2(d+1), 0.03n\}$  units of  $\mathcal{F}'$  lie within  $cs$  of  $\beta$  and they form a peak: in the model  $w\mathcal{N}(0, s^2) + (1-w)\mathcal{U}(-3cs, 3cs)$  for the residuals from  $\beta$  of the units of  $\mathcal{F}'$  within  $3cs$ , the fitted  $w$  is at least  $\frac{1}{2}$
- 4: **for**  $j = 1, \dots, K$  **do**
- 5:    $\theta^{(j)} \leftarrow$  the lines of  $\hat{\theta}$  without line  $j$ , together with  $\beta$ , refitted by the reweighting step
- 6: **end for**
- 7: score every fit by the log-likelihood  $\ell$  of  $K$  Gaussian lines plus a uniform noise component of density  $\pi_0/R$ , with the scales, proportions and noise weight  $\pi_0$  read off the fit (the units of the initial covariate screen count as noise)
- 8: among the  $\theta^{(j)}$  whose scales are all at most  $\sqrt{12} \min_k s_k$ , take the one with the largest  $\ell$ ; it replaces  $\hat{\theta}$  if  $\ell(\theta^{(j)}) - \ell(\hat{\theta})$  exceeds the margin
- 9: apply the step again while the fit changes, at most three times in all

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The steps have simple reasons. A missing group of the model lies close to one line, so a line is sought among the flagged units, as in RANSAC (Fischler & Bolles, 1981). Gross outliers also contain lines that pass near many of them, so a candidate must stand out from its surroundings: most flagged units near it must sit in a peak of the width of the fitted groups, not in a band. Which of the  $K$  original lines the candidate replaces is left to the data. The comparison uses the likelihood of the model that the flagging rule assumes: Gaussian groups, and flagged units spread over the range of the response. Under this model, two lines through one group describe it hardly better than one line, so giving up one of them costs little, whereas a group of  $n_1$  units described as noise instead of by its own line loses about  $n_1\{\log(\pi_1 R/\pi_0) - \log \sigma_1 - 1.42\}$  in log-likelihood (Proposition 3). The margin is the Bayesian information criterion's cost of the one line that was searched for, and the restriction on the scales is that of TCLUST-REG: it prevents a candidate fit from absorbing the contamination into a wide group. The step is not attempted when a third of the units or more are flagged, because the fit of Algorithm 1 is then itself in doubt. The constants of Algorithm 2 are fixed, not fitted to the data: the cut-off  $c$  and the group scales of Algorithm 1; the TCLUST-REG restriction factor 12; the majority  $\frac{1}{2}$  for a peak and its window of  $3c$  scales; the minimum support  $\max\{2(d+1), 0.03n\}$ ; the noise range 1.1 times the range of the response; the limit of one third of flagged units; 500 random lines; and at most three applications. We call the combination of the two algorithms *the procedure*.

## 2.4 Tuning constants and cost

Besides  $K$ , Algorithm 1 has six constants, and two more in the covariate screen; the constants of Algorithm 2 were given above. We used the same values in every experiment.

- The subsample size  $m$  must allow every group  $d + 1$  units, so  $m \geq (d + 1)/\pi_{\min}$ . Larger values make clean replicates more accurate but rarer: a subsample of size  $m$  is clean with probability close to  $(1 - \varepsilon)^m$  when a fraction  $\varepsilon$  of the data are outliers. We used  $m = 8$  for two groups and one covariate,  $m = 10$  when one group has 30% of the units, and  $m = 12$  for three groups or three covariates. The results change little once  $m$  exceeds  $(d + 1)/\pi_{\min}$ : from that value up to  $m = 16$  or 20 the mean accuracies in four designs varied by at most 0.02, whereas smaller values lost up to 0.06 when one group was small (Supplement, Table S12). In practice,  $m$  can therefore be set from a lower bound on the smallest group proportion, with room to spare. Since this uses knowledge that the competitors are not given, we also ran the first two parts of the study with  $m = 12$  in every design (Supplement, Table S15). Up to 20% of outliers the mean accuracies moved by at most 0.02, and the worst was 0.86, in D5; from 25% on they fell by up to 0.04, because larger subsamples are less often clean.
- The number of replicates is  $B = 100$ , drawn in  $L = 10$  stages of 10. With ten replicates per stage, the first stage contains a clean replicate with probability  $1 - \{1 - (1 - \varepsilon)^m\}^{10}$ , which is 0.84 for  $m = 8$  and  $\varepsilon = 0.2$ ; later stages do better, because flagged outliers are no longer drawn.
- The flagging cut-off is  $c = 2.5$ , the usual value for reweighting in robust regression (Rousseeuw & Leroy, 1987); at the true parameter and without overlap between groups, a clean unit is flagged with probability  $2\Phi(-2.5) \approx 0.012$ , because the scale of a group then equals its noise scale (Theorem 1).
- The cap of the bounded loss is  $c_b = 3$ , slightly above  $c$  so that a unit at the flagging boundary still contributes its full residual to the score.
- The vote weight is  $\lambda = 3$ . With  $\lambda = 0$  every replicate votes equally, and as  $\lambda \rightarrow \infty$  only the best replicate votes. In a preliminary study the second choice gave accuracies within 0.015 of  $\lambda = 3$ , whereas equal votes lost up to 0.05 when one group was small (Supplement).
- The covariate screen flags units beyond the 0.975 quantile of  $\chi_p^2$ , the usual choice for reweighting the minimum covariance determinant, and is not applied to groups with fewer than  $\max(10, 5p)$  units, whose robust distances are unstable.

The results are not sensitive to  $c$ : between  $c = 2$  and  $c = 3$  the mean accuracies changed by about 0.04 at most in the four designs of Supplement Table S3 (D1, D3, D5 and D6). The quantile of the covariate screen matters more when there are high-leverage points: at 20% of them, the 0.95 quantile gave an accuracy of 0.91 against 0.84 for 0.975 and 0.70 for 0.99, and differences of at most 0.015 in the designs without such points. We kept the pre-registered 0.975.

One run of Algorithm 1 costs  $B$  exact subsample fits (plus ten for the pilot and any redraws),  $B$  assignment problems of size  $K$  and  $O(BnK)$  operations to score the replicates. For  $n = 300$  this took between 0.03 and 0.7 seconds with one covariate in our implementation, and about two seconds with three covariates, most of it in the covariate screen. Algorithm 2 added a median of 0.01 seconds. The code is available at <https://github.com/sorujov/eesp>.

### 3 Theoretical results

This section gives three partial results. The first concerns one stage of the sequential draw and holds conditionally on the data; it explains why clean replicates are found and why the flagged set helps. The second concerns the final reweighting step and describes its limit in large samples. The third values the two operations of the recovery step in the population. None covers the passage from one stage to the next, the weighted vote, or the search for the candidate line, and we do not claim a result for them. Proofs are in Appendix A.

Throughout,  $O \subseteq \{1, \dots, n\}$  is a fixed set of contaminated units whose values are arbitrary, and a subsample is *clean* if it contains no unit of  $O$ .

#### 3.1 One stage of the sequential draw

Let  $\mathcal{F}_{\ell-1}$  be the flagged set after stage  $\ell - 1$ , with  $\mathcal{F}_0$  the covariate screen, and let  $E_\ell = \{1, \dots, n\} \setminus \mathcal{F}_{\ell-1}$  be the set of units from which stage  $\ell$  draws. (If fewer than  $m + 1$  units remain,  $E_\ell$  is the whole sample; this never happened in our experiments.) Write  $N_\ell = |E_\ell|$  for the number of eligible units and  $\varepsilon_\ell = |E_\ell \cap O|/N_\ell$  for the fraction of them that are contaminated. Let  $\mathcal{H}_{\ell-1}$  denote the data together with everything drawn before stage  $\ell$ .

Each replicate of stage  $\ell$  is obtained by drawing a uniform subsample of size  $m$  from  $E_\ell$  until an admissible one is found. Admissibility depends only on the subsample and the data. Let  $q_\ell^c$  be the fraction of admissible subsamples among the clean subsamples of  $E_\ell$ , and  $q_\ell > 0$  the fraction of admissible subsamples among all subsamples of  $E_\ell$ .

**Lemma 1.** *Conditionally on  $\mathcal{H}_{\ell-1}$ , the subsamples of stage  $\ell$  are independent, and each is uniform on the admissible subsamples of size  $m$  of  $E_\ell$ . In particular two subsamples of the same stage are conditionally independent however many units they share.*

**Proposition 2.** *Let  $p_\ell = \binom{N_\ell - |E_\ell \cap O|}{m} / \binom{N_\ell}{m}$  be the probability that a uniform subsample of  $E_\ell$  is clean, and  $a_\ell = q_\ell^c / q_\ell$ . Conditionally on  $\mathcal{H}_{\ell-1}$ , each replicate of stage  $\ell$  is clean with probability  $\pi_\ell = p_\ell a_\ell$ , and*

$$\mathbb{P}(\text{stage } \ell \text{ contains a clean replicate} \mid \mathcal{H}_{\ell-1}) = 1 - (1 - p_\ell a_\ell)^{B/L}.$$

Moreover  $a_\ell \geq q_\ell^c$  and, when  $|E_\ell \cap O| \leq N_\ell - m$ ,

$$\left(1 - \frac{\varepsilon_\ell N_\ell}{N_\ell - m + 1}\right)^m \leq p_\ell \leq (1 - \varepsilon_\ell)^m.$$

The proposition is exact. It says that what matters for stage  $\ell$  is the contamination  $\varepsilon_\ell$  among the units still eligible, not the contamination of the sample. The flagged set lowers  $\varepsilon_\ell$  when it contains outliers, and the probability of a clean replicate rises geometrically in  $m$  as it does so. With  $m = 8$ , a stage of ten replicates contains a clean one with probability 0.84 when  $\varepsilon_\ell = 0.2$  and essentially one when  $\varepsilon_\ell = 0.05$ , taking  $a_\ell = 1$ . The proposition does not say that earlier stages lower  $\varepsilon_\ell$ ; that depends on which units they flagged.

The same conditional independence gives a guarantee for a plurality vote within one stage; it is stated and proved in Appendix A. It requires  $(1 - \varepsilon_\ell)^m$  to exceed one half, and so applies only at stages where flagging has already removed most of the outliers.

### 3.2 The final reweighting step

The last step of Algorithm 1 takes a parameter  $\theta$ , recomputes the flags and refits each group on its unflagged units. We study this step on its own, at a fixed input and as  $n \rightarrow \infty$ . Let the units be independent with law  $P = (1 - \varepsilon)P_0 + \varepsilon C$ , where under the clean law  $P_0$  a unit has label  $k$  with probability  $\pi_k > 0$ , covariates with finite second moment, and response  $y = x^\top \theta_k^0 + \sigma_k e$  with  $e \sim \mathcal{N}(0, 1)$  independent of the covariates. The contaminating law  $C$  is arbitrary, except that every contaminated unit lies beyond  $c' > c$  scales from the nearest surface (Assumption 2 in Appendix A). Write  $\eta$  for the probability that a clean unit is nearer to another true surface than to its own;  $\eta$  measures the overlap of the groups.

**Theorem 1.** *Apply the reweighting step at the true parameter  $\theta = \theta^0$ . Under the conditions above and mild regularity conditions (Appendix A), almost surely as  $n \rightarrow \infty$ :*

- (i) *the scale of group  $k$  converges to a limit  $s_k$  with  $s_k/\sigma_k$  within  $O(\eta + \varepsilon)$  of one, with explicit bounds;*
- (ii) *the flagged fraction converges to  $\varepsilon + (1 - \varepsilon)\kappa$ , where  $\kappa$ , the fraction of clean units flagged, lies within  $\eta$  of  $\sum_k \pi_k 2\Phi(-cs_k/\sigma_k)$ ;*
- (iii) *the refitted coefficients converge to a limit, an average over the clean units that depends on the contamination only through the scales  $s_k$ , which lies within  $2cs_k M_x \eta / (\pi_k \lambda_k)$  of  $\theta_k^0$  when the covariates are bounded by  $M_x$  and  $\lambda_k > 0$  is the constant defined in the appendix.*

When the groups do not overlap and there is no contamination,  $\theta^0$  is a fixed point of the step and the flagged fraction tends to  $2\Phi(-c)$ .

The theorem justifies reading the flagged fraction as an estimate of the contamination: it estimates  $\varepsilon$  plus the fraction  $\kappa$  of clean units flagged, which is close to  $2\Phi(-c) \approx 0.012$  when the groups overlap little. It also shows where the bias of the final fit comes from. The contamination is removed entirely once it lies beyond the cut-off; what remains is a bias of order  $\eta$ , caused by the clean units of one group that lie nearer to another surface. The theorem assumes that the input of the step is the true parameter, or, in the general version of Appendix A, a fixed parameter; it does not show that the procedure delivers such an input.

### 3.3 The group-recovery step

The recovery step scores a fit  $F$ , a set of lines  $\theta_k$  with scales  $s_k$  and proportions  $\pi_k$  and a noise weight  $\pi_0 > 0$  summing to one, by  $\ell_n(F) = \sum_{i \leq n} \log f_F(u_i)$ , where

$$f_F(u) = \sum_k \frac{\pi_k}{s_k} \varphi\left(\frac{y - x^\top \theta_k}{s_k}\right) + \frac{\pi_0}{R}$$

and  $R$  is the range constant of the noise component. The step reads the parameters off the data; here we take them as given and evaluate the score in the population,  $L(F) = \mathbb{E}_P \log f_F(u)$ . The summands of  $\ell_n(F)$  are bounded, so for independent units  $\ell_n(F)/n \rightarrow L(F)$  almost surely. A candidate differs from the original by two operations: a line is added for a group left in the noise, and one of two lines that shared a group is removed. We value each.

**Proposition 3.** *Let the units have law  $P$ .*

(i) Let  $G$  be an event with  $P(G) = \pi_*$ , on which  $y = x^\top \theta_* + \sigma_* e$  with  $e \sim \mathcal{N}(0, 1)$  independent of  $x$  given  $G$ . Let  $F$  have noise weight  $\pi_0 > \pi_*$ , and let  $F'$  be  $F$  with the line  $(\theta_*, \sigma_*, \pi_*)$  added and the noise weight lowered to  $\pi_0 - \pi_*$ . Then

$$L(F') - L(F) \geq \pi_* \left\{ \log \frac{\pi_* R}{\pi_0} - \log \sigma_* - \frac{1}{2} \log(2\pi e) - \xi \right\} - \nu \log \frac{\pi_0}{\pi_0 - \pi_*}, \quad (3)$$

with  $\xi = \mathbb{E}[\log\{(R/\pi_0)f_F(u)\} \mid G] \geq 0$ , the extent to which  $F$  already explains  $G$ , and  $\nu = \mathbb{E}[(\pi_0/R)f_F(u)^{-1}\mathbf{1}_{G^c}] \leq 1 - \pi_*$ , the noise mass that  $F$  attributes to the units outside  $G$ . Moreover  $\xi \leq \log\{1 + (R/\pi_0) \sum_k \pi_k \mathbb{E}[\varphi_k(x^\top(\theta_* - \theta_k)) \mid G]\}$ , where  $\varphi_k$  is the centred normal density of variance  $s_k^2 + \sigma_*^2$ .

(ii) Let  $S$  be an event with  $P(S) = \pi_S$ , on which  $y = x^\top \theta_S + \sigma_S e$  as in (i). Let  $F$  contain the lines  $\theta_S \pm \frac{\delta}{2} e_1$ ,  $e_1$  the intercept coordinate, each with scale  $\sigma_S$  and proportion  $\pi_S/2$ , and let  $F''$  replace them by the line  $(\theta_S, \sigma_S, \pi_S)$ , all else equal. With  $a = \delta/(2\sigma_S)$ ,

$$-\varphi(1) a^2 \pi_S \leq L(F'') - L(F) \leq (\varphi(1) a^2 + \frac{1}{4} a^4) \pi_S,$$

and if the other components are scaled by  $t \downarrow 0$  the difference tends to  $\pi_S$  times the Kullback–Leibler divergence from  $\mathcal{N}(0, 1)$  to  $\frac{1}{2}\mathcal{N}(-a, 1) + \frac{1}{2}\mathcal{N}(a, 1)$ , which lies in  $[0, a^4/4]$ .

The leading term of (3) is the value of a group: per unit, the log of the ratio of its peak density  $\pi_*/\sigma_*$  to the noise density  $\pi_0/R$ , less the entropy  $\frac{1}{2} \log(2\pi e) \approx 1.42$  of a unit normal error; the proof (Appendix A.4) shows it is exact up to  $\xi$ , up to a nonnegative term that vanishes as the noise floor falls below the group's density, and up to the density of the other lines on  $G$ . The correction  $\xi$  is small once the group lies a few scales from every line. The loss is paid by the units that  $F$  describes as noise; when  $\nu$  is the contamination mass  $\pi_0 - \pi_*$  it is below  $\pi_*$  nats in all. With  $R/\sigma_* = 40$ ,  $\pi_* = 0.1$ ,  $\pi_0 = 0.2$  and  $\xi \approx 0$ , the value is 1.58 nats per unit of the group, that is 0.158n in all, and the loss at most  $0.1 \log 2 \approx 0.069n$ , a gain of at least  $0.09n$  against a margin  $\frac{d+1}{2} \log n$  of 8.6 for  $d = 2$ ,  $n = 300$ . Part (ii) says that removing one of two lines through one group is nearly free: the change is  $O((\delta/\sigma_S)^2)$  per unit of  $S$ , and a gain of order  $(\delta/\sigma_S)^4$  when the other components are negligible on  $S$ . Hence, if a candidate  $F_1$  arises from the original  $F_0$  by the two operations and the right side of (3) exceeds  $\varphi(1) a^2 \pi_S$ , then almost surely  $\ell_n(F_1) - \ell_n(F_0) > \frac{d+1}{2} \log n$  for all large  $n$ , and the candidate is accepted if its scales are admissible. This is for fixed parameters; the step reads them off the data, and the reweighting refit of the candidate is not covered.

The local peak test guards against a slice of a band of gross outliers, which the likelihood would value like a group. Appendix A.4 shows that, for a residual density fixed in advance, the test is a concave maximisation whose solution is zero for a band that fills the window and equals the Gaussian share for a Gaussian group over a uniform background.

## 4 Simulation study

### 4.1 Design

**Data.** Each data set has  $n = 300$  units. The covariate is uniform on  $[-3, 3]$  and the errors are standard normal. Table 2 lists the eight designs. In D1, D3, D4 and D6 the two lines cross at the origin, with slopes  $\pm 1.5$ , so the groups overlap near the crossing and even the Bayes classifier with known parameters labels about 9% of the units wrongly; an accuracy of about 0.91 is therefore the ceiling there. In D8 the lines are parallel and far apart, and the ceiling is essentially one.

Table 2: Simulation designs. All have  $n = 300$  and a uniform covariate on  $[-3, 3]$ .

| Design | Groups                                         | Outliers                    | $m$ |
|--------|------------------------------------------------|-----------------------------|-----|
| D1     | two crossing lines, equal weights and noise    | in the response             | 8   |
| D2     | three parallel lines, three noise scales apart | in the response             | 12  |
| D3     | as D1, weights 0.7 and 0.3                     | in the response             | 10  |
| D4     | as D1, noise scales 0.5 and 1.5                | in the response             | 8   |
| D5     | as D1                                          | high-leverage points        | 8   |
| D6     | as D1                                          | uniform over the data range | 8   |
| D7     | as D1, two further covariates with slope 1     | in the response             | 12  |
| D8     | two parallel lines six noise scales apart      | in the response             | 8   |

**Outliers.** Each unit is replaced by an outlier independently with probability  $\varepsilon$ . Outliers in the response have their response shifted up or down by a uniform amount between 15 and 30 times the average noise scale. High-leverage outliers (D5) have their covariate redrawn uniformly on  $[6, 9]$ , outside the design range, and their response uniformly on  $[-3, 3]$ , far from both lines there. Uniform outliers (D6) are spread over the bounding box of the clean data enlarged by 20%, and any point within three noise scales of a true line is redrawn, so that every outlier is a genuine one.

**Competitors.** Five robust methods were run from their authors’ software:

- TCLUST-REG (García-Escudero et al., 2010), function `tclustreg` of the FSDA toolbox (Riani et al., 2012), with restriction factor 12 and 300 random starts, at trimming levels  $\alpha \in \{0.05, 0.10, 0.15, 0.25, 0.40\}$ ; in D5 and D6 also at  $\alpha \in \{0.10, 0.25\}$  with 5% second trimming in the covariates;
- TCLUST-REG at each level followed by the reweighting step of Algorithm 1; this carries to clusterwise regression the idea that Dotto et al. (2018) proposed for robust clustering, but the step is ours;
- the trimmed cluster-weighted model (García-Escudero et al., 2017) and the trimmed likelihood estimator (Neykov et al., 2007), at the same five levels;
- component-wise adaptive trimming, CTLE (Chang et al., 2020), which needs no level;
- the bisquare mixture of regressions (Bai et al., 2012), which uses a bounded loss instead of trimming.

We added three methods after the pre-registered analyses of Gate 1, because they answer the question of the level in other ways, by modelling the outliers instead of trimming them. The first is the mixture of regressions with Laplace errors of Song et al. (2014), function `mixLp` of `RobMixReg` with 20 random starts, which replaces the Gaussian likelihood by a heavy-tailed one and neither trims nor flags. The second is the mixture of contaminated normal regressions (Punzo & McNicholas, 2017; Mazza & Punzo, 2020): in each group a proportion of at least one half of good points has the group’s noise scale, and the rest, the bad points, have an inflated scale; both proportions and inflations are estimated, and a unit is flagged when its posterior probability of being bad exceeds one half. We fitted it by an ECM algorithm from 50 starts, each from  $K$  random elemental lines with the units far from all of them starting as bad points, and bounded the ratio of the group variances by 12; starts from random partitions, which we tried first, were captured by the outliers. The third is a mixture of  $K$  Gaussian regressions with an additional component uniform over the range of the response enlarged by 10%, in the manner of Banfield

& Raftery (1993). Its weight is estimated with the other parameters, and a unit is flagged when its posterior probability of belonging to the uniform component exceeds one half. We fitted it by EM from 10 random partitions, continued the best after 10 iterations, and bounded the ratio of the component variances by 12, as in TCLUST-REG. No package for clusterwise regression offers these two models, so the implementations are ours; they are in the repository. This gives 25 configurations (29 in D5 and D6). The trimmed cluster-weighted model, TLE, CTLE, the bisquare mixture and the Laplace mixture come from the R package `RobMixReg` (Cao et al., 2026); the trimmed cluster-weighted model was given 50 random starts and 20 concentration steps, more than its defaults. FSDA is a MATLAB toolbox; we ran it unmodified (GitHub commit `37f998e` of 24 September 2026) under GNU Octave 9.4 with the `statistics` package (1.6.7 for the simulations, 1.6.3 for the data of Section 5), with three one-line substitutes for MATLAB functions that Octave lacks: a version check that returns false, `coder.target`, which returns true for "MATLAB", and `convertStringsToChars`, which returns its arguments; they are in the repository. The R methods used `RobMixReg` 1.1.3. We had no MATLAB licence and could not check that MATLAB gives the same results. Least squares with 50 random starts is included as a reference without protection. A simple trimmed alternation of our own at the same five levels, with the reweighting step, completes the set that the pre-registered comparison used; it is a home-made baseline, so it is reported in the Supplement, although it is the best competitor in some cells of Table S1. The reweighted versions of TCLUST-REG at a high level are the natural automatic alternative to our procedure: trim generously, then recover the units that the fit explains. The monitoring approach of Torti et al. (2019) was not run, because it leaves the final choice of level to the analyst.

**Criterion.** Every method is given the true number of groups  $K$ . Each method returns  $K$  coefficient vectors, from which every clean unit receives the label of its nearest fitted surface. The *accuracy* is the fraction of clean units whose label agrees with the true one, maximised over relabellings. The same rule is applied to every method, so trimmed units and flagged units are treated alike. A failed fit counts as accuracy zero. Failures were rare except for CTLE, which failed on 36 of the 1,600 data sets of the first part used below and on 154 of the 2,400 of the second; counted over its successful fits only, its worst mean accuracy up to 20% of outliers is still 0.59 in D2 and 0.72 in D8.

**Pre-registration.** The study has three parts and three sets of criteria, each written down and dated before the data it concerns were analysed. The first part covers  $\varepsilon \in \{0, 0.05, 0.10, 0.20, 0.30\}$  with 50 data sets per design and level, 2,000 in all; its criteria (Gate 1) concerned Algorithm 1 at  $\varepsilon \leq 0.20$ . The second part was added after the first and covers  $\varepsilon \in \{0.25, 0.30, 0.35\}$  on new data, with 100 data sets per design and level, 2,400 in all (Gate 1b). A third part, descriptive, varies the sample size, the number and size of the groups, the covariates, the errors and the type of outlier (2,300 data sets; Section 4.5). It showed that Algorithm 1 takes small groups for contamination. Algorithm 2 was then developed on data generated with other random seeds, and its criteria (Gate 2) were fixed before it was applied to any of the 6,400 data sets that the three parts then held (Section 4.6). Below, *the procedure* is Algorithm 1 followed by Algorithm 2; Algorithm 1 alone is reported as an ablation. All comparisons are paired: every method sees the same data sets.

## 4.2 Up to 20% of outliers

Table 3 shows, for each design and method, the lowest mean accuracy over  $\varepsilon \in \{0, 0.05, 0.10, 0.20\}$ . If a method works only at the right trimming level, at least one entry in its row is low.

Table 3: Worst mean accuracy on clean units over  $\varepsilon \in \{0, 0.05, 0.10, 0.20\}$ , by design. Values below 0.85 in italics. Columns: the seven designs without high-leverage points, then D5. \*With 5% second trimming in the covariates (run in D5 and D6 only).

| Method                                          | D1          | D2          | D3          | D4          | D6          | D7          | D8          | D5          |
|-------------------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Proposed procedure (Algorithms 1 and 2)         | 0.91        | 0.89        | 0.90        | 0.90        | 0.89        | 0.91        | 1.00        | 0.85        |
| Algorithm 1 alone (ablation)                    | 0.91        | 0.89        | 0.89        | 0.90        | 0.90        | 0.91        | 1.00        | 0.85        |
| Earlier version, no covariate screen (ablation) | 0.90        | 0.89        | 0.90        | 0.90        | 0.87        | 0.90        | 1.00        | <i>0.53</i> |
| TCLUST-REG, $\alpha = 0.05$                     | <i>0.53</i> | <i>0.37</i> | <i>0.69</i> | <i>0.55</i> | 0.86        | <i>0.52</i> | <i>0.52</i> | <i>0.54</i> |
| TCLUST-REG, $\alpha = 0.10$                     | <i>0.53</i> | <i>0.46</i> | <i>0.70</i> | <i>0.52</i> | 0.91        | <i>0.53</i> | <i>0.53</i> | <i>0.55</i> |
| TCLUST-REG, $\alpha = 0.15$                     | <i>0.54</i> | <i>0.60</i> | <i>0.78</i> | <i>0.65</i> | 0.91        | <i>0.55</i> | <i>0.58</i> | <i>0.57</i> |
| TCLUST-REG, $\alpha = 0.25$                     | 0.91        | <i>0.84</i> | 0.89        | 0.91        | 0.91        | 0.90        | 1.00        | <i>0.83</i> |
| TCLUST-REG, $\alpha = 0.40$                     | 0.90        | <i>0.76</i> | <i>0.68</i> | 0.90        | 0.91        | 0.90        | 1.00        | 0.87        |
| TCLUST-REG, $\alpha = 0.25$ , reweighted        | 0.91        | 0.87        | 0.89        | 0.90        | 0.91        | 0.91        | 1.00        | <i>0.83</i> |
| TCLUST-REG, $\alpha = 0.40$ , reweighted        | 0.90        | <i>0.80</i> | <i>0.67</i> | 0.90        | 0.91        | 0.91        | 1.00        | 0.87        |
| TCLUST-REG*, $\alpha = 0.10$ , reweighted       | —           | —           | —           | —           | 0.91        | —           | —           | <i>0.57</i> |
| TCLUST-REG*, $\alpha = 0.25$ , reweighted       | —           | —           | —           | —           | 0.91        | —           | —           | 0.86        |
| Trimmed CWM, $\alpha = 0.05$                    | <i>0.56</i> | <i>0.43</i> | <i>0.70</i> | <i>0.56</i> | <i>0.82</i> | <i>0.54</i> | <i>0.58</i> | <i>0.67</i> |
| Trimmed CWM, $\alpha = 0.10$                    | <i>0.54</i> | <i>0.51</i> | <i>0.71</i> | <i>0.54</i> | <i>0.85</i> | <i>0.54</i> | <i>0.58</i> | <i>0.68</i> |
| Trimmed CWM, $\alpha = 0.15$                    | <i>0.58</i> | <i>0.54</i> | <i>0.77</i> | <i>0.63</i> | 0.87        | <i>0.56</i> | <i>0.67</i> | <i>0.70</i> |
| Trimmed CWM, $\alpha = 0.25$                    | 0.87        | <i>0.65</i> | <i>0.61</i> | <i>0.84</i> | 0.86        | <i>0.81</i> | <i>0.80</i> | <i>0.80</i> |
| Trimmed CWM, $\alpha = 0.40$                    | <i>0.74</i> | <i>0.62</i> | <i>0.59</i> | <i>0.73</i> | <i>0.78</i> | <i>0.81</i> | 0.88        | <i>0.76</i> |
| TLE, $\alpha = 0.05$                            | <i>0.55</i> | <i>0.46</i> | <i>0.83</i> | <i>0.58</i> | 0.89        | <i>0.53</i> | <i>0.54</i> | <i>0.53</i> |
| TLE, $\alpha = 0.10$                            | <i>0.53</i> | <i>0.55</i> | <i>0.82</i> | <i>0.53</i> | 0.90        | <i>0.54</i> | <i>0.55</i> | <i>0.53</i> |
| TLE, $\alpha = 0.15$                            | <i>0.56</i> | <i>0.57</i> | 0.88        | <i>0.65</i> | 0.91        | <i>0.59</i> | <i>0.54</i> | <i>0.53</i> |
| TLE, $\alpha = 0.25$                            | 0.90        | <i>0.72</i> | <i>0.66</i> | 0.91        | 0.91        | 0.89        | <i>0.80</i> | <i>0.53</i> |
| TLE, $\alpha = 0.40$                            | 0.90        | <i>0.67</i> | <i>0.58</i> | 0.89        | 0.86        | <i>0.83</i> | 0.93        | <i>0.53</i> |
| CTLE                                            | 0.91        | <i>0.59</i> | 0.87        | 0.85        | 0.87        | 0.89        | <i>0.72</i> | <i>0.73</i> |
| Bisquare mixture                                | 0.89        | <i>0.56</i> | 0.88        | 0.90        | 0.90        | 0.90        | <i>0.80</i> | <i>0.55</i> |
| Laplace mixture                                 | 0.91        | <i>0.61</i> | 0.91        | 0.91        | 0.91        | 0.91        | 0.96        | <i>0.58</i> |
| Contaminated normal mixture                     | 0.91        | <i>0.52</i> | 0.91        | 0.91        | 0.91        | 0.91        | 1.00        | <i>0.83</i> |
| Noise-component mixture                         | 0.91        | <i>0.75</i> | 0.91        | 0.90        | 0.91        | 0.90        | <i>0.81</i> | <i>0.53</i> |
| Least squares (50 starts)                       | <i>0.53</i> | <i>0.37</i> | <i>0.70</i> | <i>0.52</i> | <i>0.58</i> | <i>0.53</i> | <i>0.53</i> | <i>0.53</i> |

The procedure and Algorithm 1 alone are the only methods whose rows stay at or above 0.85 in every design, although in D5 the margin (their 0.85 against 0.83 for TCLUST-REG at level 0.25 with reweighting, and 0.86 for the same with second trimming, which was run in D5 and D6 only) is within the variation between random seeds. Each fixed-level method fails somewhere, and in a predictable way. A level below the contamination breaks down: at  $\alpha \leq 0.15$ , TCLUST-REG, the trimmed cluster-weighted model and TLE drop to accuracies of about 0.7 or less at  $\varepsilon = 0.20$  in every design except D3 and D6, and to 0.6 or less in D2. A high level discards genuine units: at  $\alpha = 0.40$  each of them loses the smaller group in D3, where it holds 30% of the units, and loses accuracy in D2, where three groups share the data. The five methods without a level all fail with three groups (D2), and all but the Laplace and contaminated normal mixtures also fail in the well-separated design D8.

The pre-registered comparison of Gate 1 was stricter. It used the seven designs without high-leverage points and the four levels  $\varepsilon \leq 0.20$ , that is 28 cells. In each cell we took the competitor with the highest mean accuracy, chosen after the run, and compared it with Algorithm 1 on the same data sets. The criterion was that no cell show a mean paired difference below  $-0.03$ , nor an upper 95% bound (difference plus 1.96 standard errors) below  $-0.02$ . The largest shortfall of Algorithm 1 was 0.029 (standard error 0.014), in D3 at  $\varepsilon = 0.20$ , and in 23 of the 28 cells it was below 0.005 (Supplement, Table S1); no cell failed. Repeated for the procedure as part of Gate 2, the comparison gives a largest shortfall of 0.017 (standard error 0.011), in D6 at  $\varepsilon = 0.20$ , and again 23 cells below 0.005. When the three mixtures added afterwards are included among the competitors, the largest shortfall of the procedure is unchanged, and that of Algorithm 1 alone is 0.030, in the same cell, against the noise-component mixture. Shortfalls of this size are of the order of the variation between random seeds: in D3 at  $\varepsilon = 0.20$ , eight runs of Algorithm 1 with different seeds on the same 50 data sets gave mean accuracies between 0.885 and 0.914 (Supplement, Tables S3 and S12, and the repository).

The second criterion of Gate 1 asked that every fixed-level competitor fall at least 0.05 below Algorithm 1 in some cell. It failed for one competitor: TCLUST-REG at the conservative level 0.25 followed by reweighting was never more than 0.033 below Algorithm 1 (D2,  $\varepsilon = 0$ ). We therefore tested this competitor again, on new data, at higher contamination.

### 4.3 Beyond 25% of outliers

The second part of the study (Gate 1b) is confirmatory: it was designed after the first, to test the one competitor that the first did not separate from Algorithm 1. It asked whether TCLUST-REG at level 0.25 with reweighting fails once the contamination exceeds its level while Algorithm 1 does not. The criterion required a gap of at least 0.05, with a 95% interval excluding zero, in at least one design without high-leverage points. It was met by a wide margin. At  $\varepsilon = 0.30$  and 0.35 this competitor fell to accuracies between 0.53 and 0.62 in D1, D4, D7 and D8, where Algorithm 1 stayed above 0.87; the gap was between 0.29 and 0.47. Already at  $\varepsilon = 0.25$  it trailed by 0.06 in D1 and D8, 0.10 in D7 and 0.12 in D2, all with intervals excluding zero. The procedure gives the same picture: the recovery step is not attempted when a third of the units or more are flagged, and it changed the mean accuracy of these cells by at most 0.013.

Table 4 and Figure 1 put the two parts together. Neither conservative level works in every design. Level 0.25 breaks down as soon as the outliers outnumber it, and level 0.40, which survives high contamination, loses the smaller group in D3 and part of the third group in D2 when there are few outliers (Table 3). CTLE and the bisquare mixture fall below 0.85 at some  $\varepsilon \geq 0.25$  in every design. So does the noise-component mixture, except in D6, where its uniform component is the model that generated the outliers and it keeps an accuracy of 0.87; elsewhere its accuracy is below

Table 4: Worst mean accuracy on clean units over  $\varepsilon \in \{0.25, 0.30, 0.35\}$  (second part of the study, new data), by design. Values below 0.85 in italics. Columns: the seven designs without high-leverage points, then D5. \*With 5% second trimming in the covariates (run in D5 and D6 only).

| Method                                          | D1          | D2          | D3          | D4          | D6          | D7          | D8          | D5          |
|-------------------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Proposed procedure (Algorithms 1 and 2)         | 0.91        | <i>0.55</i> | <i>0.84</i> | 0.90        | <i>0.58</i> | 0.88        | 1.00        | <i>0.53</i> |
| Algorithm 1 alone (ablation)                    | 0.91        | <i>0.56</i> | <i>0.84</i> | 0.90        | <i>0.59</i> | 0.88        | 1.00        | <i>0.53</i> |
| Earlier version, no covariate screen (ablation) | 0.87        | <i>0.52</i> | <i>0.81</i> | 0.88        | <i>0.56</i> | <i>0.75</i> | 1.00        | <i>0.53</i> |
| TCLUST-REG, $\alpha = 0.05$                     | <i>0.54</i> | <i>0.37</i> | <i>0.68</i> | <i>0.54</i> | <i>0.56</i> | <i>0.53</i> | <i>0.53</i> | <i>0.53</i> |
| TCLUST-REG, $\alpha = 0.10$                     | <i>0.53</i> | <i>0.37</i> | <i>0.69</i> | <i>0.53</i> | <i>0.58</i> | <i>0.53</i> | <i>0.53</i> | <i>0.53</i> |
| TCLUST-REG, $\alpha = 0.15$                     | <i>0.53</i> | <i>0.38</i> | <i>0.70</i> | <i>0.53</i> | <i>0.64</i> | <i>0.53</i> | <i>0.53</i> | <i>0.53</i> |
| TCLUST-REG, $\alpha = 0.25$                     | <i>0.53</i> | <i>0.54</i> | <i>0.71</i> | <i>0.53</i> | <i>0.79</i> | <i>0.53</i> | <i>0.53</i> | <i>0.53</i> |
| TCLUST-REG, $\alpha = 0.40$                     | 0.91        | <i>0.73</i> | 0.91        | 0.91        | 0.90        | 0.86        | 1.00        | <i>0.55</i> |
| TCLUST-REG, $\alpha = 0.25$ , reweighted        | <i>0.53</i> | <i>0.54</i> | <i>0.70</i> | <i>0.53</i> | <i>0.78</i> | <i>0.53</i> | <i>0.53</i> | <i>0.53</i> |
| TCLUST-REG, $\alpha = 0.40$ , reweighted        | 0.91        | <i>0.70</i> | 0.87        | 0.90        | 0.90        | 0.86        | 1.00        | <i>0.55</i> |
| TCLUST-REG*, $\alpha = 0.10$ , reweighted       | —           | —           | —           | —           | <i>0.58</i> | —           | —           | <i>0.53</i> |
| TCLUST-REG*, $\alpha = 0.25$ , reweighted       | —           | —           | —           | —           | <i>0.80</i> | —           | —           | <i>0.53</i> |
| Trimmed CWM, $\alpha = 0.05$                    | <i>0.58</i> | <i>0.41</i> | <i>0.68</i> | <i>0.58</i> | <i>0.67</i> | <i>0.55</i> | <i>0.61</i> | <i>0.63</i> |
| Trimmed CWM, $\alpha = 0.10$                    | <i>0.55</i> | <i>0.41</i> | <i>0.68</i> | <i>0.55</i> | <i>0.65</i> | <i>0.54</i> | <i>0.56</i> | <i>0.65</i> |
| Trimmed CWM, $\alpha = 0.15$                    | <i>0.54</i> | <i>0.44</i> | <i>0.70</i> | <i>0.54</i> | <i>0.67</i> | <i>0.54</i> | <i>0.56</i> | <i>0.67</i> |
| Trimmed CWM, $\alpha = 0.25$                    | <i>0.55</i> | <i>0.53</i> | <i>0.71</i> | <i>0.55</i> | <i>0.68</i> | <i>0.54</i> | <i>0.56</i> | <i>0.67</i> |
| Trimmed CWM, $\alpha = 0.40$                    | <i>0.85</i> | <i>0.62</i> | <i>0.64</i> | <i>0.83</i> | <i>0.72</i> | <i>0.68</i> | <i>0.80</i> | <i>0.70</i> |
| TLE, $\alpha = 0.05$                            | <i>0.58</i> | <i>0.38</i> | <i>0.81</i> | <i>0.60</i> | <i>0.73</i> | <i>0.55</i> | <i>0.61</i> | <i>0.53</i> |
| TLE, $\alpha = 0.10$                            | <i>0.54</i> | <i>0.39</i> | <i>0.77</i> | <i>0.55</i> | <i>0.80</i> | <i>0.53</i> | <i>0.54</i> | <i>0.53</i> |
| TLE, $\alpha = 0.15$                            | <i>0.53</i> | <i>0.39</i> | <i>0.76</i> | <i>0.55</i> | <i>0.81</i> | <i>0.53</i> | <i>0.53</i> | <i>0.53</i> |
| TLE, $\alpha = 0.25$                            | <i>0.53</i> | <i>0.50</i> | <i>0.80</i> | <i>0.56</i> | 0.89        | <i>0.54</i> | <i>0.54</i> | <i>0.53</i> |
| TLE, $\alpha = 0.40$                            | 0.91        | <i>0.71</i> | <i>0.68</i> | 0.91        | 0.88        | 0.89        | 0.86        | <i>0.53</i> |
| CTLE                                            | <i>0.65</i> | <i>0.40</i> | <i>0.65</i> | <i>0.47</i> | <i>0.67</i> | <i>0.55</i> | <i>0.66</i> | <i>0.51</i> |
| Bisquare mixture                                | <i>0.73</i> | <i>0.42</i> | <i>0.82</i> | <i>0.76</i> | <i>0.83</i> | <i>0.63</i> | <i>0.75</i> | <i>0.53</i> |
| Laplace mixture                                 | 0.91        | <i>0.48</i> | 0.90        | 0.91        | 0.90        | 0.90        | 0.94        | <i>0.53</i> |
| Contaminated normal mixture                     | 0.90        | <i>0.37</i> | 0.91        | 0.90        | <i>0.67</i> | <i>0.65</i> | 0.85        | <i>0.61</i> |
| Noise-component mixture                         | <i>0.55</i> | <i>0.40</i> | <i>0.72</i> | <i>0.55</i> | 0.87        | <i>0.54</i> | <i>0.53</i> | <i>0.52</i> |
| Least squares (50 starts)                       | <i>0.53</i> | <i>0.37</i> | <i>0.70</i> | <i>0.53</i> | <i>0.53</i> | <i>0.53</i> | <i>0.53</i> | <i>0.53</i> |

0.80 at  $\varepsilon = 0.30$ , and in D1, D2, D4, D7 and D8 its flagged fraction falls short of  $\varepsilon$ , because part of the outliers is fitted by one of the regression components. The contaminated normal mixture keeps 0.90 or more at every level in D1, D3 and D4, but falls below 0.85 from 25% in D7 and from 30% in D6, and with three groups it fails already at 10% (0.72). The Laplace mixture is the strongest competitor at high contamination. In the six designs with two groups and without high-leverage points it keeps an accuracy of 0.90 or more up to  $\varepsilon = 0.35$ , above the procedure in D3 and D6, and in D7 at  $\varepsilon = 0.35$ . It fails, however, with three groups, already without outliers (0.66 at  $\varepsilon = 0$ ; on six data sets we checked, 100 random starts instead of 20 did not recover the three lines), and with high-leverage points (0.58 at  $\varepsilon = 0.20$ ), and it gives no estimate of the contamination.

The same tables show where the procedure itself breaks down.

- With uniform background noise (D6) its accuracy falls to 0.83 at  $\varepsilon = 0.25$  and 0.58 at 0.35, while TCLUST-REG at level 0.40 and the Laplace mixture stay at 0.90 and the noise-component mixture at 0.87.
- With three groups (D2) it falls to 0.76 at  $\varepsilon = 0.30$  and 0.55 at 0.35; TCLUST-REG at level 0.40 is not clearly better at 0.30 (0.78) but is at 0.35 (0.73).
- With high-leverage outliers (D5) it falls to 0.64 at  $\varepsilon = 0.25$  and about 0.53 beyond. No method is reliable there; the best is the trimmed cluster-weighted model at level 0.40, which models the covariates and reaches 0.83 at  $\varepsilon = 0.25$  and 0.70 at 0.35.
- With groups of 70% and 30% (D3) it holds up to  $\varepsilon = 0.30$ ; at 0.35 it reaches 0.84, and TCLUST-REG and TLE at level 0.40 and the Laplace mixture are ahead by 0.06 to 0.07.

In D1, D4, D7 and D8 it holds up to 35% of outliers. The flagged fraction signals the breakdown: where the procedure fails,  $\hat{\alpha}$  falls well below  $\varepsilon$  (Table 5), because the outliers have been absorbed into a fitted line and are no longer flagged.

#### 4.4 High-leverage outliers, the estimated contamination, and cost

The covariate screen is what makes the procedure robust to high-leverage points. Without it (third row of Table 3) Algorithm 1 breaks down in D5 at  $\varepsilon = 0.20$ , as least squares does; with it the worst accuracy up to 20% of outliers is 0.85, against 0.87 for TCLUST-REG at level 0.40 and 0.83 at level 0.25 with reweighting; TCLUST-REG with second trimming in the covariates and reweighting reaches 0.86 (Supplement, Table S11). The third row of Tables 3 and 4 is an ablation: an earlier version of Algorithm 1, without the screen and with replicates scored on all units. Apart from D5, the two versions differ by at most 0.03 in every cell up to  $\varepsilon = 0.20$  (the screen costs 0.03 in D3 at  $\varepsilon = 0.20$  and gains 0.03 in D6); from  $\varepsilon = 0.25$  on it gains up to 0.13, in D2 and D7 especially, and costs at most 0.02.

Table 5 reports the mean flagged fraction  $\hat{\alpha}$  of the procedure, and Supplement Table S16 the composition of the flagged set: where the procedure works it flags at least 91% of the outliers, in most cells all of them, and at most 3% of the clean units. Wherever the procedure works it lies within about three percentage points of  $\varepsilon$ . At low contamination it exceeds  $\varepsilon$  slightly, because of the clean units flagged (Theorem 1); the excess is largest in D4, where the noisier group overlaps the other most. In D2 without outliers it is 0.006, half the value  $2\Phi(-2.5) \approx 0.012$  of Theorem 1. The three lines there are only three noise scales apart, so no clean unit is more than 1.5 noise scales from a line except in the outer tails of the two outer groups, and only those can be flagged. The flagged fraction falls well below  $\varepsilon$  where the procedure breaks down: from  $\varepsilon = 0.25$  in D5 and D6 and from  $\varepsilon = 0.30$  in D2.

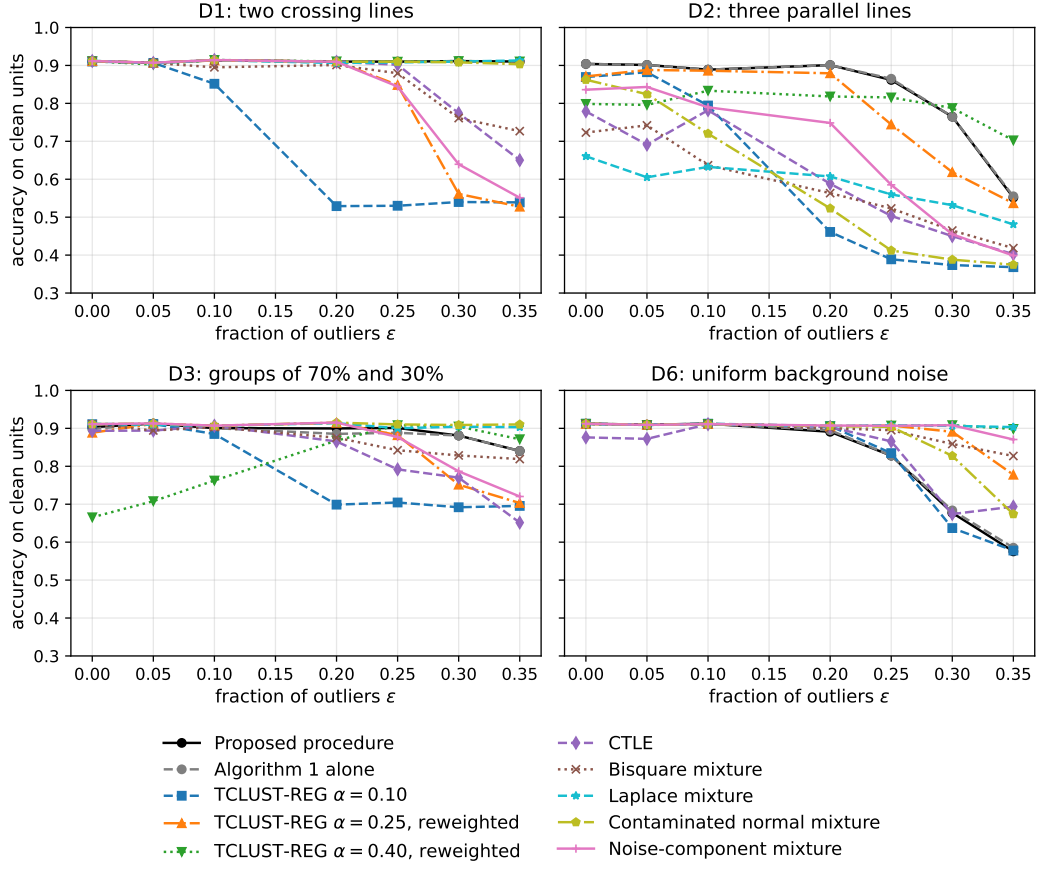


Figure 1: Mean accuracy on clean units against the fraction of outliers in four designs. Points at  $\varepsilon \leq 0.20$  are from the first part of the study (50 data sets each), points at  $\varepsilon \geq 0.25$  from the second (100 data sets each, new data).

Table 5: Mean flagged fraction  $\hat{\alpha}$  of the procedure, by design and fraction of outliers  $\varepsilon$ .

| Design | 0.00  | 0.05  | 0.10  | 0.20  | 0.25  | 0.30  | 0.35  |
|--------|-------|-------|-------|-------|-------|-------|-------|
| D1     | 0.019 | 0.060 | 0.106 | 0.211 | 0.248 | 0.306 | 0.347 |
| D2     | 0.006 | 0.050 | 0.101 | 0.199 | 0.230 | 0.213 | 0.108 |
| D3     | 0.020 | 0.065 | 0.112 | 0.201 | 0.248 | 0.293 | 0.302 |
| D4     | 0.033 | 0.076 | 0.118 | 0.207 | 0.257 | 0.304 | 0.347 |
| D6     | 0.018 | 0.058 | 0.106 | 0.170 | 0.151 | 0.074 | 0.018 |
| D7     | 0.028 | 0.068 | 0.115 | 0.208 | 0.257 | 0.304 | 0.344 |
| D8     | 0.014 | 0.057 | 0.105 | 0.204 | 0.250 | 0.301 | 0.350 |
| D5     | 0.019 | 0.062 | 0.111 | 0.182 | 0.119 | 0.051 | 0.021 |

Table 6: Computing time per data set in seconds, the first two parts of the study pooled (without the  $\varepsilon = 0.30$  cells of the first part, which the second part replaces). TCLUS-REG ran under GNU Octave, which is slower than MATLAB, so its times overstate its cost and are indicative only.

| Method                      | Median | 90th percentile |
|-----------------------------|--------|-----------------|
| Proposed procedure          | 0.08   | 1.78            |
| Algorithm 1 alone           | 0.07   | 1.66            |
| TCLUS-REG                   | 3.32   | 5.11            |
| TCLUS-REG, second trimming  | 176.44 | 240.00          |
| Trimmed CWM                 | 4.05   | 5.42            |
| TLE                         | 10.42  | 27.16           |
| CTLE                        | 14.63  | 36.71           |
| Bisquare mixture            | 3.76   | 9.13            |
| Laplace mixture             | 4.46   | 9.65            |
| Contaminated normal mixture | 0.46   | 0.76            |
| Noise-component mixture     | 0.02   | 0.04            |

**Does the exact fit matter?** We replaced the exact solution of each subsample in Algorithm 1 by a heuristic and kept everything else (Supplement, Table S14). With two groups of equal size and one covariate the choice hardly matters. Elsewhere it does. One start of hard alternation on the subsample loses up to 0.04 in D2 and D7 and 0.11 with four groups. Elemental fits, in which  $d$  units drawn for each group are fitted exactly as in RANSAC, lose up to 0.11 in D2, D3 and D7 and 0.25 with four groups. Ten starts of alternation come closest, but still lose 0.02 to 0.04 with four groups. Pooled over the designs with one covariate, the exact fit was also faster than alternation (median 0.03 against 0.05 to 0.10 seconds per data set); with four groups it was slower (1.5 against 0.1 to 0.3 seconds).

Table 6 gives computing times on one core. The procedure took a median of 0.08 seconds per data set, of which Algorithm 2 took about 0.01, and about two seconds with three covariates, where most of the time goes to the minimum covariance determinant of the covariate screen. The published competitors took medians of two to fifteen seconds, and about three minutes with second trimming; the noise-component mixture, a plain EM fit, took 0.02 seconds. When the level is unknown, a fixed-level method must moreover be run at several levels.

#### 4.5 Sample size, groups, covariates and outliers

The third, descriptive part of the study (2,300 data sets, 50 per cell) varied what the first two held fixed. It used

- design D1 with  $n = 100$  and  $n = 1000$ , and design D3 with  $n = 1000$ ;
- four parallel lines four noise scales apart, with  $n = 400$  and  $m = 16$ ;
- design D7 with the three covariates dependent through a Gaussian copula with correlation 0.5;
- designs D1 and D3 with moderate outliers, whose response is shifted by 4 to 8 noise scales instead of 15 to 30;
- design D1 with groups of 80% and 20% ( $m = 16$ ) and of 85% and 15% ( $m = 20$ );
- design D1 with the outliers concentrated in a tight cloud, at covariate value about 2 and response about 20 noise scales;

- designs D1 and D3 with Student  $t$  errors on three degrees of freedom, scaled to unit variance, at  $\varepsilon \in \{0, 0.10, 0.20\}$  (added after Gate 2).

Table 7 summarises it.

The sample size changes little. With  $n = 100$  and  $n = 1000$  the procedure stays within 0.013 of the best competitor chosen afterwards in every cell, up to  $\varepsilon = 0.30$ , and its running time grows from 0.07 to about 0.1 seconds, against two to about thirty seconds for the published competitors at  $n = 1000$ . With four groups and outliers it is clearly the best method: at  $\varepsilon = 0.10$  and 0.20 its accuracy is 0.96 and 0.95, against 0.89 and 0.79 for the best competitor, and TCLUST-REG at the conservative levels stays below 0.84 even without outliers. It breaks down at  $\varepsilon = 0.30$  (0.72), where no method does better than 0.75. Dependent covariates make no difference to the accuracy, although the covariate screen then flags two to three percentage points more of the clean units than with independent covariates, because the joint distribution of the covariates is not elliptical.

Moderate outliers are harder to flag, because many of them lie within a few noise scales of the other line. The flagged fraction then falls short of  $\varepsilon$ , by up to thirteen percentage points at  $\varepsilon = 0.30$ , but the accuracy on clean units is hardly affected: the procedure stays within 0.01 of the best competitor everywhere except D3 at  $\varepsilon = 0.30$ , where it trails TCLUST-REG at level 0.25 by 0.04.

Small groups are what Algorithm 2 is for. Algorithm 1 alone reaches only 0.82 without outliers and 0.73 at  $\varepsilon = 0.20$  with groups of 80% and 20%, and between 0.60 and 0.73 at every  $\varepsilon$  with groups of 85% and 15%. Its failed fits have the same form: two nearly parallel lines that share the large group, with most of the small group flagged. The bounded loss of Algorithm 1 allows this: a unit of the small group costs at most  $c_b^2$  squared scales, and two lines fit the large group more tightly than one. A larger cap is no cure. With  $c_b = 5$  the small groups are largely recovered, but at  $\varepsilon = 0.35$  the accuracy falls from 0.91 to 0.82 in D1 and from 0.88 to 0.74 in D7, and at  $\varepsilon = 0.20$  from 0.90 to 0.79 in D2 and from 0.85 to 0.72 in D5 (Supplement, Table S13). Algorithm 2 leaves  $c_b$  alone and asks instead whether the flagged units contain a group. With it, the procedure reaches 0.88 to 0.91 with either small group up to  $\varepsilon = 0.20$ , within 0.035 of the best competitor in every such cell. At  $\varepsilon = 0.30$  the first step flags about a third of the units, the recovery step changed the fit on at most two of fifty data sets, and the accuracy stays at 0.75.

Heavy-tailed errors leave the accuracy intact: with  $t_3$  errors the procedure is within 0.005 of the best competitor in all six cells. They do change what the flagged fraction measures. Without outliers it flags 6.5% of the units, the tails of the errors that lie beyond 2.5 robust scales, and with outliers it exceeds  $\varepsilon$  by two to five percentage points while still flagging every outlier (Table 7 and the repository). The flagged fraction estimates the contamination only relative to Gaussian errors.

Clustered outliers are the reverse case. At  $\varepsilon = 0.10$  the cloud is flagged and the procedure is as accurate as the best competitor. At  $\varepsilon = 0.20$ , Algorithm 1 fits the cloud with one of its two lines, and its accuracy falls to 0.70; the recovery step repairs a quarter of these fits and raises the mean to 0.78, against 0.90 for TCLUST-REG at a level above the contamination. At  $\varepsilon = 0.30$  no method exceeds 0.68.

## 4.6 The recovery step: pre-registered confirmation

Algorithm 2 was developed on data generated as in the study but with four other seeds, and frozen before Gate 2. Gate 2 applied it to the stored fit of Algorithm 1 on every one of the 6,400 data sets that the study then held, with three criteria. (a) In the two small-group designs at  $\varepsilon \leq 0.20$ , the mean gain over Algorithm 1 was to be at least 0.05 in each of the six cells. It was between 0.07 and 0.31 (Table 8). (b) In no cell of any part with  $\varepsilon \leq 0.20$  was the procedure to be worse than

Table 7: Further designs (third part of the study, descriptive, 50 data sets per cell): mean accuracy on clean units of the proposed procedure (Algorithms 1 and 2), of Algorithm 1 alone, of the best competitor chosen afterwards in each cell, and of TCLUST-REG with reweighting at the two conservative levels;  $\hat{\alpha}$  is the mean flagged fraction of the proposed procedure.

| Design                         | $\varepsilon$ | Proposed | Alg. 1 alone | Best competitor                          |       | TCR <sub>0.25</sub> +rw | TCR <sub>0.40</sub> +rw | $\hat{\alpha}$ |
|--------------------------------|---------------|----------|--------------|------------------------------------------|-------|-------------------------|-------------------------|----------------|
| D1, $n = 100$                  | 0.00          | 0.912    | 0.913        | Contaminated normal mixture              | 0.912 | 0.910                   | 0.902                   | 0.024          |
|                                | 0.10          | 0.918    | 0.918        | TCLUST-REG, $\alpha = 0.25$ , reweighted | 0.918 | 0.918                   | 0.912                   | 0.108          |
|                                | 0.20          | 0.908    | 0.901        | Contaminated normal mixture              | 0.907 | 0.906                   | 0.900                   | 0.208          |
|                                | 0.30          | 0.908    | 0.908        | Contaminated normal mixture              | 0.912 | 0.662                   | 0.911                   | 0.296          |
| D1, $n = 1000$                 | 0.00          | 0.910    | 0.910        | Laplace mixture                          | 0.910 | 0.910                   | 0.910                   | 0.018          |
|                                | 0.10          | 0.911    | 0.911        | Noise-component mixture                  | 0.911 | 0.911                   | 0.911                   | 0.106          |
|                                | 0.20          | 0.911    | 0.911        | Laplace mixture                          | 0.912 | 0.911                   | 0.911                   | 0.201          |
|                                | 0.30          | 0.902    | 0.902        | TCLUST-REG, $\alpha = 0.40$ , reweighted | 0.910 | 0.522                   | 0.910                   | 0.301          |
| D3, $n = 1000$                 | 0.00          | 0.913    | 0.913        | TCLUST-REG, $\alpha = 0.05$              | 0.913 | 0.913                   | 0.630                   | 0.017          |
|                                | 0.10          | 0.911    | 0.911        | TCLUST-REG, $\alpha = 0.10$              | 0.912 | 0.911                   | 0.810                   | 0.104          |
|                                | 0.20          | 0.903    | 0.903        | TCLUST-REG, $\alpha = 0.25$              | 0.912 | 0.911                   | 0.911                   | 0.201          |
|                                | 0.30          | 0.901    | 0.901        | Contaminated normal mixture              | 0.914 | 0.736                   | 0.913                   | 0.298          |
| four parallel lines, $n = 400$ | 0.00          | 0.958    | 0.958        | TCLUST-REG, $\alpha = 0.05$ , reweighted | 0.964 | 0.827                   | 0.746                   | 0.005          |
|                                | 0.10          | 0.965    | 0.965        | TCLUST-REG, $\alpha = 0.15$ , reweighted | 0.892 | 0.837                   | 0.746                   | 0.099          |
|                                | 0.20          | 0.949    | 0.949        | TLE, $\alpha = 0.25$                     | 0.788 | 0.765                   | 0.686                   | 0.189          |
|                                | 0.30          | 0.724    | 0.734        | TLE, $\alpha = 0.40$                     | 0.749 | 0.689                   | 0.658                   | 0.129          |
| D7, dependent covariates       | 0.00          | 0.909    | 0.909        | Noise-component mixture                  | 0.910 | 0.908                   | 0.909                   | 0.059          |
|                                | 0.10          | 0.915    | 0.915        | Noise-component mixture                  | 0.917 | 0.915                   | 0.914                   | 0.132          |
|                                | 0.20          | 0.909    | 0.909        | CTLE                                     | 0.908 | 0.907                   | 0.907                   | 0.230          |
|                                | 0.30          | 0.910    | 0.910        | TLE, $\alpha = 0.40$                     | 0.911 | 0.552                   | 0.903                   | 0.325          |
| D1, moderate outliers          | 0.05          | 0.907    | 0.907        | TCLUST-REG, $\alpha = 0.15$ , reweighted | 0.907 | 0.907                   | 0.907                   | 0.050          |
|                                | 0.10          | 0.910    | 0.910        | TCLUST-REG, $\alpha = 0.15$ , reweighted | 0.911 | 0.911                   | 0.910                   | 0.081          |
|                                | 0.20          | 0.914    | 0.914        | TCLUST-REG, $\alpha = 0.10$              | 0.916 | 0.913                   | 0.913                   | 0.143          |
|                                | 0.30          | 0.901    | 0.901        | TCLUST-REG, $\alpha = 0.25$ , reweighted | 0.910 | 0.910                   | 0.910                   | 0.193          |
| D3, moderate outliers          | 0.05          | 0.910    | 0.910        | TCLUST-REG, $\alpha = 0.15$              | 0.912 | 0.907                   | 0.717                   | 0.052          |
|                                | 0.10          | 0.910    | 0.910        | Contaminated normal mixture              | 0.911 | 0.911                   | 0.667                   | 0.077          |
|                                | 0.20          | 0.904    | 0.904        | TCLUST-REG, $\alpha = 0.10$ , reweighted | 0.906 | 0.906                   | 0.850                   | 0.143          |
|                                | 0.30          | 0.874    | 0.874        | TCLUST-REG, $\alpha = 0.25$              | 0.911 | 0.910                   | 0.906                   | 0.169          |
| D1, groups of 80% and 20%      | 0.00          | 0.910    | 0.819        | Contaminated normal mixture              | 0.912 | 0.634                   | 0.583                   | 0.020          |
|                                | 0.10          | 0.910    | 0.836        | TCLUST-REG, $\alpha = 0.15$              | 0.911 | 0.880                   | 0.602                   | 0.108          |
|                                | 0.20          | 0.882    | 0.727        | Contaminated normal mixture              | 0.915 | 0.914                   | 0.699                   | 0.203          |
|                                | 0.30          | 0.748    | 0.748        | Contaminated normal mixture              | 0.892 | 0.820                   | 0.866                   | 0.284          |
| D1, groups of 85% and 15%      | 0.00          | 0.909    | 0.628        | Contaminated normal mixture              | 0.912 | 0.596                   | 0.566                   | 0.026          |
|                                | 0.10          | 0.912    | 0.605        | TCLUST-REG, $\alpha = 0.10$              | 0.915 | 0.731                   | 0.578                   | 0.110          |
|                                | 0.20          | 0.881    | 0.619        | TCLUST-REG, $\alpha = 0.25$              | 0.907 | 0.901                   | 0.670                   | 0.203          |
|                                | 0.30          | 0.747    | 0.732        | Contaminated normal mixture              | 0.882 | 0.852                   | 0.813                   | 0.256          |
| D1, clustered outliers         | 0.00          | 0.914    | 0.914        | Bisquare mixture                         | 0.914 | 0.913                   | 0.913                   | 0.018          |
|                                | 0.10          | 0.910    | 0.910        | Contaminated normal mixture              | 0.911 | 0.911                   | 0.911                   | 0.109          |
|                                | 0.20          | 0.781    | 0.700        | TCLUST-REG, $\alpha = 0.40$ , reweighted | 0.905 | 0.897                   | 0.905                   | 0.190          |
|                                | 0.30          | 0.550    | 0.536        | TCLUST-REG, $\alpha = 0.40$              | 0.677 | 0.545                   | 0.676                   | 0.072          |
| D1, $t_3$ errors               | 0.00          | 0.927    | 0.927        | TCLUST-REG, $\alpha = 0.40$              | 0.928 | 0.928                   | 0.927                   | 0.065          |
|                                | 0.10          | 0.934    | 0.934        | Laplace mixture                          | 0.936 | 0.935                   | 0.935                   | 0.143          |
|                                | 0.20          | 0.931    | 0.931        | TCLUST-REG, $\alpha = 0.25$              | 0.932 | 0.931                   | 0.931                   | 0.226          |
| D3, $t_3$ errors               | 0.00          | 0.933    | 0.933        | TCLUST-REG, $\alpha = 0.10$              | 0.933 | 0.933                   | 0.853                   | 0.064          |
|                                | 0.10          | 0.930    | 0.930        | Noise-component mixture                  | 0.931 | 0.930                   | 0.919                   | 0.144          |
|                                | 0.20          | 0.926    | 0.926        | TCLUST-REG, $\alpha = 0.25$              | 0.932 | 0.931                   | 0.930                   | 0.221          |

Algorithm 1 by more than 0.01. The largest loss was 0.004 (D6,  $\varepsilon = 0.20$ ). (c) The comparison of Gate 1, repeated for the procedure, was to be passed; it was, as reported in Section 4.2. All three criteria were met.

Table 8: Gate 2: mean accuracy on clean units of Algorithm 1 alone and of the procedure, and the paired difference with its standard error, in the cells where the recovery step changed the mean by more than 0.005 (50 data sets per cell, 100 in the second part).

| Design             | $\varepsilon$ | Algorithm 1 alone | Procedure | Difference | SE    |
|--------------------|---------------|-------------------|-----------|------------|-------|
| D2 (part 1)        | 0.30          | 0.747             | 0.740     | -0.007     | 0.007 |
| D3                 | 0.00          | 0.898             | 0.903     | +0.005     | 0.005 |
| D3                 | 0.20          | 0.885             | 0.900     | +0.014     | 0.010 |
| D3                 | 0.25          | 0.888             | 0.901     | +0.013     | 0.006 |
| D6                 | 0.30          | 0.684             | 0.677     | -0.007     | 0.005 |
| D6                 | 0.35          | 0.585             | 0.576     | -0.010     | 0.006 |
| D1, $n = 100$      | 0.20          | 0.901             | 0.908     | +0.007     | 0.007 |
| four groups        | 0.30          | 0.734             | 0.724     | -0.010     | 0.006 |
| groups 80/20       | 0.00          | 0.819             | 0.910     | +0.091     | 0.022 |
| groups 80/20       | 0.10          | 0.836             | 0.910     | +0.074     | 0.021 |
| groups 80/20       | 0.20          | 0.727             | 0.882     | +0.155     | 0.026 |
| groups 85/15       | 0.00          | 0.628             | 0.909     | +0.281     | 0.023 |
| groups 85/15       | 0.10          | 0.605             | 0.912     | +0.308     | 0.020 |
| groups 85/15       | 0.20          | 0.619             | 0.881     | +0.262     | 0.023 |
| groups 85/15       | 0.30          | 0.732             | 0.747     | +0.015     | 0.011 |
| clustered outliers | 0.20          | 0.700             | 0.781     | +0.080     | 0.022 |
| clustered outliers | 0.30          | 0.536             | 0.550     | +0.013     | 0.009 |

The step changed the fit on 3.4% of all data sets. Outside the small-group designs it raised the accuracy in D3 by up to 0.014 and in the clustered-outlier design by 0.08, and it lowered it by 0.007 to 0.010 in four cells with 30% or 35% of outliers, where the procedure has already broken down (D2 at  $\varepsilon = 0.30$  in the first part, D6 at 0.30 and 0.35, four groups at 0.30).

**The recovery step after a trimming method.** Algorithm 2 needs only a fit and its flags, so it can follow any method that flags units. Applied after TCLUST-REG with reweighting (Supplement, Table S17), it repairs exactly the failures of a high level: at level 0.40 the accuracy with groups of 80% and 20% rises from between 0.58 and 0.70 to between 0.90 and 0.91 up to  $\varepsilon = 0.20$ , with 85% and 15% from between 0.57 and 0.67 to between 0.88 and 0.91, and in D3 without outliers from 0.67 to 0.91. It does not repair what a high level does to three or four groups (D2 stays at 0.80, four groups rise from 0.75 to 0.77), and with uniform noise at 30% and 35% it lowers the accuracy of TCLUST-REG at level 0.40 from 0.91 to 0.88 and from 0.90 to 0.74: a band of uniform noise that is dense and narrow enough passes the peak test (Section 3.3). The recovery step is thus a contribution in its own right; combined with a generous trimming level it gives a strong method for two groups, and combined with Algorithm 1 it keeps the accuracy of the latter with three and four groups.

## 5 Two data sets

### 5.1 The fishery data

The fishery data (Riani et al., 2008), distributed with FSDA, record 677 monthly import flows of one fishery product into the European Union from a third country: the quantity and the value of each flow. They are the standard example for TCLUST-REG, because flows priced differently form lines through the origin and a number of flows fit none of them. We work on the logarithmic scale, where  $\text{value} = \text{price} \times \text{quantity}$  becomes a line of slope one whose intercept is the log price, and where the spread of the errors no longer grows with the quantity. Twenty-six flows with zero quantity or value are left out, which leaves 651. We fit  $K = 2$  groups with the constants of Section 2.4 and  $m = 8$ .

Table 9: Fishery data, logarithmic scale,  $K = 2$ : estimated intercepts and slopes, with the range over random seeds in brackets when it exceeds 0.005, and the trimmed or flagged fraction.

| Method                                  | Group 1: intercept, slope            | Group 2: intercept, slope            | Trimmed or flagged |
|-----------------------------------------|--------------------------------------|--------------------------------------|--------------------|
| Procedure (20 seeds)                    | 2.18 [2.16, 2.58], 0.96 [0.95, 0.98] | 2.78 [2.78, 2.85], 0.95              | 0.03 [0.02, 0.12]  |
| Algorithm 1 alone (20 seeds)            | 2.57 [2.56, 2.59], 0.97 [0.97, 0.98] | 2.85 [2.84, 2.86], 0.95 [0.94, 0.95] | 0.11 [0.10, 0.12]  |
| TCLUST-REG, $\alpha = 0.02$ (10 seeds)  | 2.21, 0.93                           | 2.78, 0.94                           | 0.02               |
| TCLUST-REG, $\alpha = 0.04$ (10 seeds)  | 2.25, 0.92                           | 2.78, 0.94                           | 0.04               |
| TCLUST-REG, $\alpha = 0.06$ (10 seeds)  | 2.24, 0.92                           | 2.78, 0.94                           | 0.06               |
| TCLUST-REG, $\alpha = 0.08$ (10 seeds)  | 2.22 [2.22, 2.24], 0.92              | 2.78, 0.94                           | 0.08               |
| TCLUST-REG, $\alpha = 0.10$ (10 seeds)  | 2.25 [1.91, 2.37], 0.94 [0.90, 0.99] | 2.77, 0.95                           | 0.10               |
| TCLUST-REG, $\alpha = 0.15$ (10 seeds)  | 2.52 [1.81, 2.78], 0.87 [0.38, 0.95] | 2.79 [2.78, 2.84], 0.89 [0.79, 0.95] | 0.15               |
| TCLUST-REG, $\alpha = 0.20$ (10 seeds)  | 2.64 [2.56, 2.78], 0.94 [0.81, 1.07] | 2.79 [2.78, 2.80], 0.93 [0.81, 0.95] | 0.20               |
| TCLUST-REG, $\alpha = 0.25$ (10 seeds)  | 2.65 [2.55, 2.79], 0.96 [0.87, 1.01] | 2.81 [2.79, 2.87], 0.93 [0.84, 0.95] | 0.25               |
| TCLUST-REG, $\alpha = 0.30$ (10 seeds)  | 2.62 [2.55, 2.67], 0.96 [0.93, 1.03] | 2.80 [2.78, 2.82], 0.94 [0.94, 0.95] | 0.30               |
| CTLE (10 seeds)                         | 2.34, 0.94                           | 2.79, 0.94                           | 0.02               |
| Trimmed CWM, $\alpha = 0.04$ (10 seeds) | 2.57 [2.28, 2.71], 0.87 [0.72, 0.97] | 2.79 [2.78, 2.80], 0.93 [0.82, 0.94] | 0.04               |
| Trimmed CWM, $\alpha = 0.10$ (10 seeds) | 2.66 [2.56, 2.75], 0.95 [0.90, 1.00] | 2.79 [2.76, 2.85], 0.93 [0.83, 0.96] | 0.10               |
| Trimmed CWM, $\alpha = 0.25$ (10 seeds) | 2.69 [2.57, 2.78], 0.96 [0.91, 1.00] | 2.80 [2.78, 2.84], 0.95 [0.94, 0.96] | 0.25               |

Table 9 and Figure 2 show the results. Algorithm 1 alone finds two lines of slope close to one, with intercepts 2.57 and 2.85, that is prices of about 13 and 17 (value per unit of quantity, in the units of the data set), and flags 11% of the flows, most of them priced below both lines. The recovery step finds that these cheap flows lie close to a line of their own: for 19 of 20 seeds it replaces the line at 2.57 by a line with intercept 2.16, a price of about 9, keeps one line through the two dearer price levels (intercept 2.78), and flags 3% of the flows. With two lines, the procedure thus prefers a cheap group and one line for the rest to two close price levels and a cheap tail flagged as anomalous.

TCLUST-REG gives the same two answers, but which one depends on the trimming level. At levels up to 0.10 it treats the cheap flows as a group, with intercept near 2.2, as the procedure does. From level 0.20 on it separates the two dearer price levels, as Algorithm 1 alone does, and its estimates then vary noticeably from one random start to another. Between the two, at level 0.15, four of the ten starts return two lines with the same intercept and different slopes. CTLE agrees with TCLUST-REG at low levels, and the trimmed cluster-weighted model varies with its starts at every level.

The data suggest three price levels. With  $K = 3$  and  $m = 16$ , the procedure returns lines with intercepts 1.90 to 1.91, 2.50 to 2.54 and 2.82 to 2.83, slopes between 0.94 and 1.00, and 3% of the flows flagged, for 12 of 20 seeds; the other seeds give other fits. Algorithm 1 alone did not give a stable answer with  $K = 3$  for any subsample size from  $m = 12$  to  $m = 20$ . The dependence on the seed is removed by running the procedure a few times and keeping the fit of highest likelihood under the model of Algorithm 2. Among twenty disjoint sets of five seeds, this rule returned the

cheap group and the dearer line (intercepts 2.16 to 2.18 and 2.78) every time with  $K = 2$ , and the three price levels (intercepts 1.90 to 1.91, 2.49 to 2.54 and 2.81 to 2.84) every time with  $K = 3$ ; larger numbers of replicates,  $B = 500$  or  $1000$  instead of  $100$ , helped less. The Bayesian information criterion of the same model prefers  $K = 2$ . Whether the cheap flows are a genuine price level or anomalous declarations is a question for the subject-matter expert. What the example shows is that the procedure reaches, without a trimming level, the answer that TCLUST-REG gives only at the right level, and that it reports how many flows no line explains.

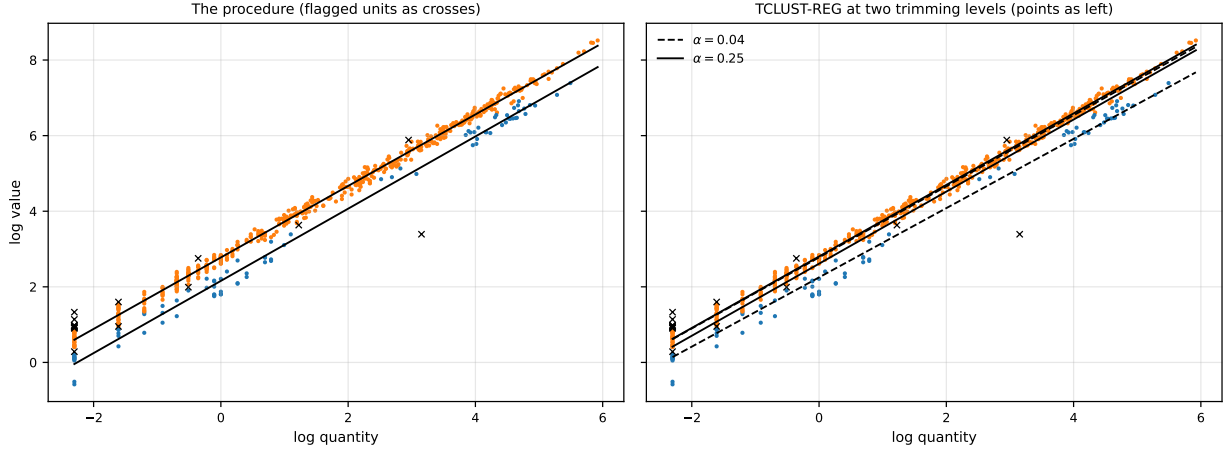


Figure 2: Fishery data on the logarithmic scale. Left: the two lines of the procedure (seed 0), with its flagged flows as crosses and the other flows coloured by group. Right: TCLUST-REG at trimming levels 0.04 (dashed) and 0.25 (solid), with the same points.

## 5.2 The tone perception data

The tone perception data (De Veaux, 1989), distributed with the R package *mixtools* (Benaglia et al., 2009), record 150 trials in which a musician tuned a tone to the octave above a fundamental tone whose overtones had been stretched by a given ratio. The tuned ratio lies either near 2, the octave, or near the stretch ratio, so the data form a nearly flat line and a line of slope one. They are a standard test for robust mixtures of regressions; Bai et al. (2012) added ten identical points at  $(0, 4)$ , far from both lines and outside the range of the covariate. We fit  $K = 2$  with the constants of Section 2.4 and  $m = 8$ , to the original data and with the ten points added (Table 10). A fit counts as “other” unless it has a nearly flat line near 2 (intercept between 1.8 and 2.1, slope below 0.1 in absolute value) and a line of slope within 0.1 of one; the fractions are over the remaining seeds, and the Laplace mixture flags no units.

On the original data every method finds the two lines, except the trimmed cluster-weighted model at level 0.25 from two of its ten starts; the rising line has intercept between 0 and 0.07 and slope between 0.97 and 1.00. The added points, 6% of the enlarged sample, separate the methods. TCLUST-REG at levels 0.02 and 0.05, the trimmed cluster-weighted model at level 0.05 and the Laplace mixture lose one of the two lines from every start, and the noise-component mixture for 7 of 20 seeds, nearly always by fitting a line through the added points; the trimmed cluster-weighted model returns other lines also from 3 of 10 starts at level 0.10 and 6 of 10 at level 0.25. The procedure returns the same two lines, to within 0.01, for every seed, as do CTLE, the contaminated normal mixture and TCLUST-REG from level 0.10 on; its recovery step found no group among the flagged trials and left the fit of Algorithm 1 unchanged.

Table 10: Tone perception data,  $K = 2$ , over 20 seeds or 10 random starts: the flat line (intercept, slope; range in brackets when it exceeds 0.005), the number of seeds with other fits, and the trimmed or flagged fraction.

| Method                       | Original data ( $n = 150$ )          |            |                   | With ten points at (0, 4) |                   |
|------------------------------|--------------------------------------|------------|-------------------|---------------------------|-------------------|
|                              | Flat line                            | Other fits | Fraction          | Other fits                | Fraction          |
| Procedure                    | 1.97, 0.02                           | 0/20       | 0.25 [0.25, 0.27] | 0/20                      | 0.29 [0.29, 0.31] |
| Contaminated normal mixture  | 1.95, 0.03                           | 0/20       | 0.23              | 0/20                      | 0.14              |
| Noise-component mixture      | 1.92, 0.04                           | 0/20       | 0.07              | 7/20                      | 0.13              |
| TCLUST-REG, $\alpha = 0.02$  | 1.93, 0.04                           | 0/10       | 0.02              | 10/10                     | –                 |
| TCLUST-REG, $\alpha = 0.05$  | 1.91, 0.04                           | 0/10       | 0.05              | 10/10                     | –                 |
| TCLUST-REG, $\alpha = 0.10$  | 1.93, 0.04                           | 0/10       | 0.10              | 0/10                      | 0.10              |
| TCLUST-REG, $\alpha = 0.15$  | 1.95, 0.03                           | 0/10       | 0.15              | 0/10                      | 0.15              |
| TCLUST-REG, $\alpha = 0.25$  | 1.98, 0.02                           | 0/10       | 0.25              | 0/10                      | 0.25              |
| TCLUST-REG, $\alpha = 0.40$  | 2.01, 0.00                           | 0/10       | 0.40              | 0/10                      | 0.40              |
| CTLE                         | 1.93, 0.04                           | 0/10       | 0.09              | 0/10                      | 0.14 [0.14, 0.14] |
| Laplace mixture              | 1.96 [1.94, 1.96], 0.02 [0.02, 0.03] | 0/10       | –                 | 10/10                     | –                 |
| Trimmed CWM, $\alpha = 0.05$ | 1.92, 0.04                           | 0/10       | 0.05              | 10/10                     | –                 |
| Trimmed CWM, $\alpha = 0.10$ | 1.93, 0.04                           | 0/10       | 0.10              | 3/10                      | 0.10              |
| Trimmed CWM, $\alpha = 0.25$ | 1.95 [1.95, 1.97], 0.03 [0.02, 0.03] | 2/10       | 0.25              | 6/10                      | 0.25              |

The flagged fraction of the procedure is 0.25 on the original data. This reflects the shape of the data, not gross outliers. Of the 85 trials nearest the rising line, about 60% lie within 0.01 of it, and the others spread up to 0.5. The robust scale of that group is set by its tight core, so a trial 0.03 from the line is already beyond the cut-off; 18 to 20 of the 85, depending on the seed, are flagged in this way. Most of the group also has stretch ratios in a narrow band near 2, and the covariate screen, which compares the stretch ratio of each trial with the median and median absolute deviation of its group, flags 14 further trials whose ratios lie outside it. The contaminated normal mixture, which also estimates a proportion of bad points, flags 23% of the original trials. Here the flagged fraction measures departures from a group with Gaussian errors and a compact covariate distribution, not contamination in the subject-matter sense; the fitted lines are not affected. With the ten points added it rises to 0.29, which is the ten points together with the trials flagged before.

## 6 Discussion

The procedure replaces the choice of a trimming level by a fixed residual cut-off, a sequence of exact fits to small subsamples, and a check of what was flagged. It finds clean subsamples, flags the units they do not fit, draws later subsamples from the rest, and finally asks whether the flagged units contain a group. In the simulations it matched the best of the competitors, chosen afterwards design by design, whenever up to a fifth of the data were outliers, while every competitor that needs a trimming level failed in some design, at low or at high contamination. Its flagged fraction is an estimate of the contamination when the errors are close to Gaussian. On the fishery data it reached, without a level, the answer that TCLUST-REG gives only at low levels, and on the tone data it was unaffected by added outliers that broke the trimming methods at low levels; there, however, the flagged fraction reflected heavy-tailed errors and a compact covariate distribution within one group rather than gross outliers.

The second step is what distinguishes a procedure that estimates its contamination from one that is merely told it. A method that decides for itself what to discard must also decide whether a poorly fitted set of units is contamination or a group, and the first step alone gets this wrong for small groups. The recovery step settles it by the likelihood of the model that the flagging rule already assumes, and its criteria were confirmed on data it had not been developed on. The same

question arises for other robust procedures that flag rather than trim, and the step can be applied after any of them.

The strongest competitors without a level were the mixtures of regressions with Laplace and with contaminated normal errors. With two groups and without high-leverage points they were about as accurate as the procedure up to 20% of outliers, and the Laplace mixture was more robust beyond; both recovered small groups. Both failed with three and with four groups, even at low contamination, the Laplace mixture also with high-leverage points, and the contaminated normal mixture from 25% or 30% of outliers in two of the two-group designs (D6 and D7). The procedure proposed here is therefore to be preferred when there may be more than two groups or outlying covariates, or when a fitted line that absorbs contamination must be avoided; with two groups and outliers in the response only, these mixtures are strong alternatives. So is a generous trimming level followed by the reweighting and recovery steps, which the recovery step makes robust to small groups.

Four limits should be kept in mind.

- *High contamination.* The procedure breaks down when outliers are so frequent that clean subsamples are rare in the first stages and a contaminated fit is taken for the reference. In our designs this happened from 25% of outliers with uniform background noise or high-leverage points and from 30% with three groups, while with two groups of equal size, or with three covariates, it held up to 35%. A flagged fraction well below any plausible contamination is the warning sign. The recovery step is not attempted when a third of the units or more are flagged, so small groups are recovered only up to about 20% of outliers.
- *Clustered outliers.* Outliers concentrated in a tight cloud were fitted as part of a group from 20% on; the recovery step repaired a quarter of these fits.
- *Heteroscedastic errors.* The flagging rule compares residuals with a scale per group. If the error spread grows with the covariate, as in raw trade data, units with large covariate values are flagged as outliers. A transformation that stabilises the spread, such as the logarithm used in Section 5, should precede the fit.
- *Theory.* Section 3 explains one stage, the final step and the value of the recovery comparison in the population, but not the passage between stages, the weighted vote or the search for the candidate line. A result for the whole procedure would need to control which units are flagged, and we do not have one.

The number of groups  $K$  was taken as known. The likelihood that the recovery step uses also gives a way to choose it: fit the procedure for several values of  $K$  and compare the Bayesian information criterion of the Gaussian-lines-plus-noise model. In a descriptive check on part of the study (Supplement, Table S18) this chose the true  $K$  for at least 90% of the data sets in every design with two or four groups and up to 20% of outliers, but for only 50% to 80% with the three heavily overlapping groups of D2, and it counted a tight cloud of outliers as a third group. On the fishery data it prefers  $K = 2$ .

On data without outliers the procedure offers nothing over least squares with random restarts: at  $\varepsilon = 0$ , least squares with 50 starts was as accurate to within 0.001 in every design, more accurate by 0.007 in D3, and faster (a median of 0.01 seconds against 0.08). The procedure is meant for data that may contain outliers in unknown proportion.

## Declarations

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**Competing interests.** The author declares no competing interests.

**Data and code availability.** The code, the scripts that produce every table and figure, and the raw simulation output are available at <https://github.com/sorujov/eesp> and archived at Zenodo (<https://doi.org/10.5281/zenodo.22807276>). The fishery data are distributed with the FSDA toolbox (Riani et al., 2012).

**Pre-registration.** The criteria of Sections 4.2 and 4.3 were written down and dated before the corresponding simulations were run; the documents are in the repository.

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## A Proofs

### A.1 Proposition 1

For fixed  $\theta$  the sum  $\sum_{i \in S} (y_i - x_i^\top \theta_{z_i})^2$  separates over the units, so its minimum over  $z$  is  $R_S(\theta) = \sum_{i \in S} \min_k (y_i - x_i^\top \theta_k)^2$ , attained by the nearest-surface labels. If  $(\hat{z}^S, \hat{\theta}^S)$  attains the joint minimum, then

$$\sum_{i \in S} (y_i - x_i^\top \hat{\theta}_{\hat{z}_i^S}^S)^2 \geq R_S(\hat{\theta}^S) \geq \min_{\theta} R_S(\theta) = \min_{z, \theta} \sum_{i \in S} (y_i - x_i^\top \theta_{z_i})^2.$$

The two ends coincide, so  $\hat{\theta}^S$  minimises  $R_S$  and each  $\hat{z}_i^S$  is a nearest surface. The converse is the same chain read with equalities.  $\square$

### A.2 Lemma 1 and Proposition 2

*Lemma 1.* Fix a replicate of stage  $\ell$ . Conditionally on  $\mathcal{H}_{\ell-1}$  its proposals are independent uniform subsamples of size  $m$  of  $E_\ell$ , because  $E_\ell$  is determined by  $\mathcal{H}_{\ell-1}$  and the draws use fresh randomness. (The implementation draws  $m$  units without replacement with equal weights on  $E_\ell$  and zero weight elsewhere, which gives a uniform subsample of  $E_\ell$ .) Whether a proposal is admissible is determined by the proposal and the data. The first admissible proposal in a sequence of independent proposals has the law of one proposal conditioned on admissibility, which is the uniform law on the admissible subsamples; the sequence stops almost surely because  $q_\ell > 0$ . Different replicates use disjoint randomness, so they are conditionally independent.  $\square$

*Proposition 2.* By the lemma, a replicate of stage  $\ell$  is clean with conditional probability equal to the number of admissible clean subsamples divided by the number of admissible subsamples. Dividing both by  $\binom{N_\ell}{m}$  gives  $\pi_\ell = p_\ell q_\ell^c / q_\ell = p_\ell a_\ell$ . The replicates of the stage are conditionally independent, which gives the probability that at least one is clean. Next,  $a_\ell \geq q_\ell^c$  because  $q_\ell \leq 1$ . Finally  $p_\ell = \prod_{j=0}^{m-1} (N_\ell - |E_\ell \cap O| - j) / (N_\ell - j)$ ; each factor equals  $1 - |E_\ell \cap O| / (N_\ell - j)$ , which lies between  $1 - |E_\ell \cap O| / (N_\ell - m + 1)$  and  $1 - |E_\ell \cap O| / N_\ell = 1 - \varepsilon_\ell$ .  $\square$

*A vote within one stage.* The same conditional independence gives a guarantee for a vote within one stage. Suppose a clean replicate gives unit  $i$  its correct label with probability at least  $\bar{q}$ , where the label refers to the least-squares partition of the clean eligible units. If  $\pi_\ell \bar{q} > \frac{1}{2}$ , a plurality vote of the replicates of stage  $\ell$  labels every such unit correctly with conditional probability at least  $1 - n \exp\{-2(B/L)(\pi_\ell \bar{q} - \frac{1}{2})^2\}$ , whatever the contaminated replicates do (proved below). Since  $\pi_\ell \approx (1 - \varepsilon_\ell)^m$ , this requires

$$m < \frac{\log(2\bar{q})}{-\log(1 - \varepsilon_\ell)} \approx \frac{\log(2\bar{q})}{\varepsilon_\ell}, \quad (4)$$

while admissibility and the accuracy of a clean replicate require  $m$  of order at least  $(d+1)/\pi_{\min}$ . The subsample size must lie between the two bounds. As the contamination among the eligible units grows, the range of admissible  $m$  shrinks; each round of flagging lowers that contamination and so widens the range again. The window can be empty: with  $\bar{q}$  close to one, (4) requires  $\varepsilon_\ell < 1 - 2^{-1/m}$ , which is 0.083 for  $m = 8$  and 0.056 for  $m = 12$ , so at the contamination levels of our experiments the guarantee applies only at stages where flagging has already removed most of the outliers. This guarantee covers a single stage, not the weighted vote across stages that Algorithm 1 uses; it motivates the design but says nothing about the final output.

*The vote within one stage.* Let  $t_i$  be the label of unit  $i$  in the least-squares partition of the clean eligible units, labelled by the same pilot as the replicates, and suppose a clean replicate gives unit  $i$  the label  $t_i$  with conditional probability at least  $\bar{q}$ . Then a replicate of stage  $\ell$  gives unit  $i$  the label  $t_i$  with conditional probability  $Q_i \geq \pi_\ell \bar{q}$ , whatever the contaminated replicates do, since the event that the replicate is clean and labels  $i$  correctly has probability  $\pi_\ell$  times the conditional probability of a correct label given cleanness. By the lemma, the number of the  $B/L$  replicates that give unit  $i$  the label  $t_i$  is binomial with success probability  $Q_i$ . If  $Q_i \geq \frac{1}{2} + \gamma$  with  $\gamma > 0$ , the plurality differs from  $t_i$  only if at most half of the replicates vote for  $t_i$ , which by Hoeffding's inequality has probability at most  $\exp\{-2(B/L)\gamma^2\}$ . A union bound over at most  $n$  units, with  $\gamma = \pi_\ell \bar{q} - \frac{1}{2}$ , gives the statement of Section 3.1. The window (4) is  $(1 - \varepsilon_\ell)^m \bar{q} > \frac{1}{2}$  solved for  $m$ , with  $a_\ell = 1$ .  $\square$

### A.3 The final reweighting step

Theorem 1 of Section 3.2 collects parts of Theorems 2 and 3 below.

The last two lines of Algorithm 1 apply a reweighting step to the parameter produced by the concentration steps. We study the step as a map on its own, applied to a fixed parameter  $\theta = (\theta_1, \dots, \theta_K) \in \mathbb{R}^{Kd}$ , and let the sample size grow. The step, as implemented, is the following. For unit  $i$  let  $z_i(\theta) = \arg \min_k |y_i - x_i^\top \theta_k|$  be its nearest surface, the smallest index in case of a tie, and  $r_i(\theta) = \min_k |y_i - x_i^\top \theta_k|$  its absolute residual from that surface. Let  $n_k = \#\{i : z_i(\theta) = k\}$ . The scale of group  $k$  is  $\hat{s}_k = 1.4826 \hat{m}_k$ , where  $\hat{m}_k$  is a sample median of  $\{r_i(\theta) : z_i(\theta) = k\}$ , by which we mean any value between the two middle order statistics of that set; the implementation takes the square root of the median of the squared residuals, which lies between them. When  $n_k < 10$  the median over all  $n$  residuals is used instead. The flagged set and the flagged fraction are

$$\mathcal{F}_n(\theta) = \{i : r_i(\theta) > c \hat{s}_{z_i(\theta)}\}, \quad \hat{\alpha}_n(\theta) = |\mathcal{F}_n(\theta)|/n.$$

The refitted parameter  $\hat{\theta}_k$  is the least-squares fit of  $y$  on  $x$  over  $\{i : z_i(\theta) = k, i \notin \mathcal{F}_n(\theta)\}$  when this set has at least  $d$  units, and  $\hat{\theta}_k = \theta_k$  otherwise.

The population counterparts are defined for a probability measure  $P$  on  $\mathcal{U} = \mathbb{R}^d \times \mathbb{R}$  and a unit  $u = (x, y)$ :

$$\begin{aligned} h_\theta(u) &= \arg \min_k |y - x^\top \theta_k|, & r_\theta(u) &= \min_k |y - x^\top \theta_k|, & A_k(\theta) &= \{u : h_\theta(u) = k\}, \\ m_k(\theta) &= \inf\{t \geq 0 : P(r_\theta \leq t \mid A_k(\theta)) \geq \tfrac{1}{2}\}, & s_k(\theta) &= 1.4826 m_k(\theta), \\ W_k(\theta) &= A_k(\theta) \cap \{r_\theta \leq c s_k(\theta)\}, & \alpha(\theta) &= P(r_\theta > c s_{h_\theta(\theta)}(\theta)) = 1 - \sum_k P(W_k(\theta)), \\ M_k(\theta) &= \mathbb{E}[xx^\top \mathbf{1}_{W_k(\theta)}], & v_k(\theta) &= \mathbb{E}[xy \mathbf{1}_{W_k(\theta)}], & \theta_k^\infty(\theta) &= M_k(\theta)^{-1} v_k(\theta). \end{aligned}$$

Here  $m_k(\theta)$  is the median of the residual  $r_\theta$  among the units assigned to surface  $k$ ,  $s_k(\theta)$  the corresponding scale,  $W_k(\theta)$  the set of units assigned to  $k$  and not flagged,  $\alpha(\theta)$  the population flagged fraction, and  $\theta^\infty(\theta)$  the population refit. The dependence on  $\theta$  is dropped when  $\theta$  is fixed.

**Assumption 1** (Clean law).  $P = (1 - \varepsilon)P_0 + \varepsilon C$  with  $\varepsilon \in [0, 1]$ . Under  $P_0$  a unit is generated by a label  $z \in \{1, \dots, K\}$  with  $\mathbb{P}(z = k) = \pi_k > 0$ , a covariate  $x$  whose conditional law given  $z = k$  has  $\mathbb{E}_0[\|x\|^2] < \infty$ , and an error  $e \sim \mathcal{N}(0, 1)$  independent of  $(x, z)$ , through  $y = x^\top \theta_z^0 + \sigma_z e$  with  $\sigma_k > 0$ . The contaminating law  $C$  is arbitrary.

**Assumption 2** (Contamination beyond the cut-off). At the given  $\theta$ , there is  $c' > c$  such that  $C(r_\theta \geq c' s_{h_\theta}(\theta)) = 1$ : every contaminated unit lies beyond  $c'$  scales from every surface of  $\theta$ , the scale being that of the surface nearest to it.

The scale  $s_k(\theta)$  in Assumption 2 is computed under  $P$ , so the assumption is a condition on  $P$ ; it is the form of “beyond the cut-off” that refers to the scale the step actually uses.

**Assumption 3** (Regularity at  $\theta$ ). For every  $k$ :  $P(A_k(\theta)) > 0$ ; the median of  $r_\theta$  on  $A_k(\theta)$  is positive and unique, that is,  $m_k(\theta) > 0$ ,  $P(r_\theta \leq t \mid A_k) < \frac{1}{2}$  for  $t < m_k$  and  $> \frac{1}{2}$  for  $t > m_k$ ; and  $M_k(\theta)$  is nonsingular.

**Lemma 2** (Plug-in cut-off). Let  $u_1, u_2, \dots$  be independent with law  $P$ , let  $k$  be fixed, let  $\hat{s}_k$  be random with  $\hat{s}_k \rightarrow s_k$  almost surely, and let  $g : \mathcal{U} \rightarrow \mathbb{R}^p$  be measurable with  $P(A_k, r_\theta = cs_k) = 0$  and  $\mathbb{E}[\|g\| \mathbf{1}\{A_k, r_\theta \leq cs_k + \delta_0\}] < \infty$  for some  $\delta_0 > 0$ . Then, almost surely,

$$\frac{1}{n} \sum_{i=1}^n g(u_i) \mathbf{1}\{z_i(\theta) = k, r_i(\theta) \leq c\hat{s}_k\} \rightarrow \mathbb{E}[g \mathbf{1}_{W_k}].$$

*Proof.* Fix  $\delta \in (0, \delta_0)$ . Almost surely  $|c\hat{s}_k - cs_k| \leq \delta$  for all large  $n$ , and then the average on the left differs from  $n^{-1} \sum_i g(u_i) \mathbf{1}\{z_i = k, r_i \leq cs_k\}$  by at most  $n^{-1} \sum_i \|g(u_i)\| \mathbf{1}\{z_i = k, cs_k - \delta < r_i \leq cs_k + \delta\}$  in norm. By the strong law of large numbers the second average converges to  $\mathbb{E}[g \mathbf{1}_{W_k}]$  and the bound to  $\mathbb{E}[\|g\| \mathbf{1}\{A_k, cs_k - \delta < r_\theta \leq cs_k + \delta\}]$ , both integrable by hypothesis. Letting  $\delta \downarrow 0$  along a sequence, the last expectation tends to  $\mathbb{E}[\|g\| \mathbf{1}\{A_k, r_\theta = cs_k\}] = 0$  by dominated convergence.  $\square$

**Theorem 2** (One reweighting step at a fixed input). Let  $u_1, \dots, u_n$  be independent with law  $P$ , fix  $\theta$ , and let Assumptions 1–3 hold at  $\theta$ . Then, almost surely as  $n \rightarrow \infty$ :

- (i)  $\hat{s}_k \rightarrow s_k(\theta)$  for every  $k$ ;
- (ii)  $\hat{\alpha}_n(\theta) \rightarrow \alpha(\theta) = \varepsilon + (1 - \varepsilon)\kappa(\theta)$ , where  $\kappa(\theta) = P_0(r_\theta > cs_{h_\theta}(\theta))$  is the flagged fraction among clean units;
- (iii)  $\hat{\theta} \rightarrow \theta^\infty(\theta)$ , and  $\theta_k^\infty(\theta) = \mathbb{E}_0[xx^\top \mathbf{1}_{W_k}]^{-1} \mathbb{E}_0[xy \mathbf{1}_{W_k}]$ : the limit of the refit is an average over the clean law alone, and the contamination enters it only through the scales  $s_k(\theta)$  that define  $W_k$ .

*Proof. Step 1: no mass at the cut-off.* Under  $P_0$ ,  $y$  has a density conditionally on  $(x, z)$ , so  $P_0(|y - x^\top \theta_j| = t) = 0$  for every  $j$  and  $t$ , and  $P_0(r_\theta = t) \leq \sum_j P_0(|y - x^\top \theta_j| = t) = 0$ . Under  $C$ , Assumption 2 gives  $C(A_k, r_\theta < c's_k) = 0$ . Hence  $P(A_k, r_\theta = cs_k) = 0$ , and for  $\delta_0 = \frac{1}{2}(c' - c) \min_k s_k$ , which is positive by Assumption 3, and every  $g \geq 0$ ,

$$\mathbb{E}[g \mathbf{1}\{A_k, r_\theta \leq cs_k + \delta_0\}] = (1 - \varepsilon) \mathbb{E}_0[g \mathbf{1}\{A_k, r_\theta \leq cs_k + \delta_0\}]. \quad (5)$$

In particular  $P(W_k) = (1 - \varepsilon)P_0(W_k)$ ,  $M_k = (1 - \varepsilon)\mathbb{E}_0[xx^\top \mathbf{1}_{W_k}]$  and  $v_k = (1 - \varepsilon)\mathbb{E}_0[xy \mathbf{1}_{W_k}]$ ; the factors  $1 - \varepsilon$  cancel in  $\theta_k^\infty$ , which proves the second claim of (iii).

*Step 2: scales.* By the strong law,  $n_k/n \rightarrow P(A_k) > 0$ , so  $n_k \geq 10$  for all large  $n$  and the pooled fallback is inactive. For  $t \geq 0$  put  $\hat{F}_k(t) = n_k^{-1} \#\{i : z_i = k, r_i \leq t\}$  and  $F_k(t) = P(r_\theta \leq t \mid A_k)$ ; then  $\hat{F}_k(t) \rightarrow F_k(t)$  almost surely for each fixed  $t$ , as a ratio of two averages. Fix  $\delta > 0$ . By the uniqueness of the median,  $F_k(m_k - \delta) < \frac{1}{2} < F_k(m_k + \delta)$ , so for all large  $n$ ,  $\hat{F}_k(m_k - \delta) < \frac{1}{2} < \hat{F}_k(m_k + \delta)$ : fewer than half of the residuals of group  $k$  are at most  $m_k - \delta$  and more than half are at most  $m_k + \delta$ . Both middle order statistics, and hence  $\hat{m}_k$ , then lie in  $(m_k - \delta, m_k + \delta]$ . Taking  $\delta = 1/j$ ,  $j \geq 1$ , gives  $\hat{m}_k \rightarrow m_k$  almost surely, which is (i).

*Step 3: flagged fraction.* Apply Lemma 2 with  $g \equiv 1$ , the hypotheses holding by Step 1 and (i):  $n^{-1} \#\{i : z_i = k, r_i \leq c\hat{s}_k\} \rightarrow P(W_k)$  for each  $k$ . Summing over  $k$ ,  $1 - \hat{\alpha}_n(\theta) \rightarrow \sum_k P(W_k) = 1 - \alpha(\theta)$ . By Step 1,  $\sum_k P(W_k) = (1 - \varepsilon) \sum_k P_0(W_k) = (1 - \varepsilon)(1 - \kappa(\theta))$ , which rearranges to (ii).

*Step 4: refit.* Apply Lemma 2 with  $g = xx^\top$  and with  $g = xy$ ; the integrability hypothesis holds by (5), since  $\mathbb{E}_0 \|x\|^2 < \infty$  and  $\mathbb{E}_0 y^2 < \infty$  under Assumption 1. Thus  $n^{-1} \sum_i x_i x_i^\top \mathbf{1}\{z_i = k, i \notin \mathcal{F}_n\} \rightarrow M_k$  and  $n^{-1} \sum_i x_i y_i \mathbf{1}\{z_i = k, i \notin \mathcal{F}_n\} \rightarrow v_k$  almost surely. Since  $M_k$  is nonsingular,  $P(W_k) > 0$ , so the number of unflagged units of group  $k$  exceeds  $d$  for all large  $n$ , the fallback is inactive, and  $\hat{\theta}_k$  is the image of the two averages under the map  $(M, v) \mapsto M^{-1}v$ , which is continuous at  $(M_k, v_k)$ . This is (iii).  $\square$

Theorem 2 identifies the limit of the step at any input satisfying the assumptions. It does not say that the input of the step in Algorithm 1 is close to  $\theta^0$ . The next theorem evaluates the limits at  $\theta = \theta^0$ , where they can be made explicit, and quantifies the effect of overlap between the groups. Write

$$\eta_k = P_0(h_{\theta^0} \neq k \mid z = k), \quad \eta = \sum_k \pi_k \eta_k = P_0(h_{\theta^0} \neq z),$$

for the probability that a clean unit of group  $k$ , or a clean unit, is nearer to another true surface than to its own. Let  $p_k = P(A_k(\theta^0))$  and

$$\delta_k = 1 - \frac{(1 - \varepsilon)\pi_k(1 - \eta_k)}{p_k}, \quad \nu_k = \delta_k + \frac{1}{2}\eta_k,$$

so that  $\delta_k$  is the fraction of the units assigned to surface  $k$  that are not clean units of group  $k$ ; it satisfies  $\delta_k p_k \leq (1 - \varepsilon)\eta + \varepsilon$ , since the clean units of other groups assigned to  $k$  have mass at most  $(1 - \varepsilon)\eta$ . Finally let

$$q(u) = 1.4826 \Phi^{-1}\left(\frac{1+u}{2}\right), \quad 0 < u < 1,$$

an increasing function with  $q(\frac{1}{2}) = 1.4826 \Phi^{-1}(\frac{3}{4}) = 1.0000$  to four decimals;  $q(u)\sigma$  is the scale the step assigns to a  $\mathcal{N}(0, \sigma^2)$  residual whose  $u$ -quantile in absolute value is used in place of the median.

**Theorem 3** (The step at the true parameter). *Let Assumption 1 hold, let  $\theta = \theta^0$ , and suppose that  $P_0(x^\top(\theta_k^0 - \theta_j^0) = 0) = 0$  for  $j \neq k$  (no two surfaces coincide on a set of covariates of positive mass), that  $\nu_k < \frac{1}{2}$  for every  $k$ , and that Assumption 2 holds at  $\theta^0$ .*

(i) *For every  $k$ ,  $p_k > 0$  and the median of  $r_{\theta^0}$  on  $A_k(\theta^0)$  is positive and unique, and*

$$q\left(\frac{1}{2} - \nu_k\right) \leq \frac{s_k(\theta^0)}{\sigma_k} \leq q\left(\frac{1}{2} + \nu_k\right).$$

(ii) *The clean flagged fraction satisfies*

$$\sum_k \pi_k 2\Phi(-c q(\frac{1}{2} + \nu_k)) - \eta \leq \kappa(\theta^0) \leq \sum_k \pi_k 2\Phi(-c q(\frac{1}{2} - \nu_k)) + \eta,$$

$$\text{and } \alpha(\theta^0) = \varepsilon + (1 - \varepsilon)\kappa(\theta^0).$$

(iii) Suppose in addition that  $\|x\| \leq M_x$   $P_0$ -almost surely and that

$$\lambda_k := \{2\Phi(cq(\frac{1}{2} - \nu_k)) - 1\} \lambda_{\min}(\mathbb{E}_0[xx^\top \mid z = k]) - M_x^2 \eta_k > 0.$$

Then  $M_k(\theta^0)$  is nonsingular with  $\lambda_{\min}(M_k(\theta^0)) \geq (1 - \varepsilon)\pi_k \lambda_k$ , Assumption 3 holds at  $\theta^0$ , Theorem 2 applies, and

$$\|\theta_k^\infty(\theta^0) - \theta_k^0\| \leq \frac{2c s_k(\theta^0) M_x}{\pi_k \lambda_k} \eta.$$

In particular, if  $\varepsilon = 0$  and  $\eta = 0$  then  $s_k(\theta^0) = q(\frac{1}{2})\sigma_k$ ,  $\kappa(\theta^0) = 2\Phi(-cq(\frac{1}{2}))$  and  $\theta^\infty(\theta^0) = \theta^0$ : the true parameter is a fixed point of the population reweighting step, and the flagged fraction is  $2\Phi(-c)$  up to the rounding in 1.4826.

*Proof.* Throughout,  $h = h_{\theta^0}$ ,  $r = r_{\theta^0}$ ,  $A_k = A_k(\theta^0)$ ,  $s_k = s_k(\theta^0)$  and  $G_k(t) = \mathbb{P}(\sigma_k|e| \leq t) = 2\Phi(t/\sigma_k) - 1$ .

*Geometry of the assignment.* For  $j \neq k$  put  $\Delta_j(x) = x^\top(\theta_k^0 - \theta_j^0)$ , which is nonzero  $P_0$ -almost surely. On  $\{z = k\}$  we have  $y - x^\top \theta_k^0 = \sigma_k e$  and  $y - x^\top \theta_j^0 = \sigma_k e + \Delta_j$ , and  $|a| < |a + \Delta|$  if and only if  $\Delta(2a + \Delta) > 0$ . Hence, given  $x$ , the event  $\{h = k\}$  is  $\{-L(x) < \sigma_k e < U(x)\}$  with  $L(x) = \min\{\Delta_j/2 : \Delta_j > 0\}$  and  $U(x) = \min\{-\Delta_j/2 : \Delta_j < 0\}$ , the minimum over an empty set being  $+\infty$ ; both are positive. Ties have probability zero. On  $\{z = k, h = k\}$  the residual is  $r = \sigma_k|e|$ .

*Composition of  $A_k$ .* The mass of  $A_k$  splits into the clean units of group  $k$  with  $h = k$ , of mass  $(1 - \varepsilon)\pi_k(1 - \eta_k)$ , and the rest, of mass  $\delta_k p_k$ ; the rest consists of clean units of other groups, of mass at most  $(1 - \varepsilon)\sum_j \pi_j P_0(h \neq j \mid z = j) = (1 - \varepsilon)\eta$ , and of contaminated units. Since  $\delta_k \leq \nu_k < \frac{1}{2}$ ,  $p_k > 0$  and  $\eta_k < 1$ . The conditional distribution function of  $r$  on  $A_k$  is therefore

$$F_k(t) = (1 - \delta_k)F_k^{\text{own}}(t) + \delta_k F_k^{\text{oth}}(t), \quad F_k^{\text{own}}(t) = \frac{P_0(\sigma_k|e| \leq t, h = k \mid z = k)}{1 - \eta_k},$$

with  $F_k^{\text{oth}}$  some distribution function. Since  $e$  is independent of  $(x, z)$ ,  $P_0(\sigma_k|e| \leq t \mid z = k) = G_k(t)$ , and  $P_0(\sigma_k|e| \leq t, h = k \mid z = k)$  lies between  $G_k(t) - \eta_k$  and  $G_k(t)$ . Hence

$$(1 - \delta_k) \frac{G_k(t) - \eta_k}{1 - \eta_k} \leq F_k(t) \leq (1 - \delta_k) \frac{G_k(t)}{1 - \eta_k} + \delta_k. \quad (6)$$

(i) *Uniqueness.* By the geometry above,  $F_k^{\text{own}}$  has density

$$f_k^{\text{own}}(t) = \frac{\varphi(t/\sigma_k)}{\sigma_k(1 - \eta_k)} \mathbb{E}_0[\mathbf{1}\{t < U(x)\} + \mathbf{1}\{t < L(x)\} \mid z = k],$$

which is positive exactly when  $P_0(\max\{L(x), U(x)\} > t \mid z = k) > 0$ ; this property is monotone in  $t$ , so  $f_k^{\text{own}} > 0$  on  $[0, T_k)$  and  $f_k^{\text{own}} = 0$  on  $[T_k, \infty)$  for some  $T_k \in (0, \infty]$ . If the median of  $F_k$  were not unique,  $F_k$  would equal  $\frac{1}{2}$  on an interval of positive length. On  $[0, T_k)$  the function  $F_k$  is strictly increasing, so the interval lies in  $[T_k, \infty)$ , where  $F_k^{\text{own}} = 1$  and  $F_k \geq 1 - \delta_k > \frac{1}{2}$ , a contradiction. The median is positive because  $F_k^{\text{own}}(0) = 0$  gives  $F_k(0) \leq \delta_k < \frac{1}{2}$ .

(i) *Bounds.* Let  $m_k$  be the median. If  $F_k(t) \geq \frac{1}{2}$  then, by the upper bound in (6),  $G_k(t) \geq u_k^- := (\frac{1}{2} - \delta_k)(1 - \eta_k)/(1 - \delta_k)$ , so  $m_k \geq G_k^{-1}(u_k^-)$ . If  $G_k(t) \geq u_k^+ := \eta_k + (1 - \eta_k)/(2(1 - \delta_k))$  then, by the lower bound,  $F_k(t) \geq \frac{1}{2}$ , so  $m_k \leq G_k^{-1}(u_k^+)$ ; here  $0 < u_k^-$  and  $u_k^+ < 1$  because  $\delta_k < \frac{1}{2}$  and  $\eta_k < 1$ . Now  $(\frac{1}{2} - \delta_k)/(1 - \delta_k) \geq \frac{1}{2} - \delta_k$  and  $1/(2(1 - \delta_k)) \leq \frac{1}{2} + \delta_k$  for  $\delta_k \leq \frac{1}{2}$ , whence  $u_k^- \geq (\frac{1}{2} - \delta_k)(1 - \eta_k) \geq \frac{1}{2} - \nu_k > 0$  and  $u_k^+ \leq \eta_k + (1 - \eta_k)(\frac{1}{2} + \delta_k) \leq \frac{1}{2} + \nu_k < 1$ . Since  $G_k^{-1}(u) = \sigma_k \Phi^{-1}((1 + u)/2)$  is increasing and  $s_k = 1.4826 m_k$ , the bounds follow.

(ii). Split  $\kappa(\theta^0) = P_0(r > cs_h)$  according to whether  $h = z$ . On  $\{h = z = k\}$ ,  $r = \sigma_k|e|$  and  $s_h = s_k$ , so  $P_0(h = z, r > cs_h) = \sum_k \pi_k P_0(\sigma_k|e| > cs_k, h = k | z = k)$ , which lies between  $\sum_k \pi_k \{G'_k - \eta_k\}$  and  $\sum_k \pi_k G'_k$  with  $G'_k = \mathbb{P}(\sigma_k|e| > cs_k) = 2\Phi(-cs_k/\sigma_k)$ . The event  $\{h \neq z\}$  contributes between 0 and  $\eta$ . Inserting the bounds of (i) on  $s_k/\sigma_k$  into  $G'_k$  gives (ii); the expression for  $\alpha(\theta^0)$  is Theorem 2(ii), whose Assumption 3 is verified in (iii) below, or directly from (5).

(iii) *Nonsingularity.* By (5),  $M_k = (1 - \varepsilon)\mathbb{E}_0[xx^\top \mathbf{1}_{W_k}]$ , and  $W_k \supseteq \{z = k, h = k, \sigma_k|e| \leq cs_k\}$ . In the positive semidefinite order,

$$\begin{aligned} \mathbb{E}_0[xx^\top \mathbf{1}\{z = k, h = k, \sigma_k|e| \leq cs_k\}] \\ = \mathbb{E}_0[xx^\top \mathbf{1}\{z = k, \sigma_k|e| \leq cs_k\}] - \mathbb{E}_0[xx^\top \mathbf{1}\{z = k, h \neq k, \sigma_k|e| \leq cs_k\}], \end{aligned}$$

where the first term equals  $\pi_k G_k(cs_k)\mathbb{E}_0[xx^\top | z = k]$  by the independence of  $e$  and  $(x, z)$ , and the second is at most  $M_x^2 \pi_k \eta_k I$  because  $xx^\top \preceq \|x\|^2 I$ . With  $G_k(cs_k) = 2\Phi(cs_k/\sigma_k) - 1 \geq 2\Phi(cq(\frac{1}{2} - \nu_k)) - 1$  by (i),  $\lambda_{\min}(M_k) \geq (1 - \varepsilon)\pi_k \lambda_k > 0$ . The other two parts of Assumption 3 were shown in (i).

(iii) *Bias.*  $\theta_k^\infty - \theta_k^0 = M_k^{-1}\mathbb{E}[x(y - x^\top \theta_k^0)\mathbf{1}_{W_k}]$ , and by (5) the expectation is  $(1 - \varepsilon)$  times its  $P_0$ -version. Split the latter by  $z$ . On  $\{z = k\}$ ,  $y - x^\top \theta_k^0 = \sigma_k e$  and  $W_k \cap \{z = k\} = \{z = k, h = k, |e| \leq cs_k/\sigma_k\}$ , so

$$\begin{aligned} \mathbb{E}_0[xe \mathbf{1}\{z = k, h = k, |e| \leq cs_k/\sigma_k\}] \\ = \mathbb{E}_0[x \mathbf{1}\{z = k\}] \mathbb{E}[e \mathbf{1}\{|e| \leq cs_k/\sigma_k\}] - \mathbb{E}_0[xe \mathbf{1}\{z = k, h \neq k, |e| \leq cs_k/\sigma_k\}]. \end{aligned}$$

The first term vanishes by the symmetry of  $e$ ; the second has norm at most  $(cs_k/\sigma_k)M_x \pi_k \eta_k$ , so the contribution of  $\{z = k\}$  is at most  $cs_k M_x \pi_k \eta_k$  in norm. On  $\{z = j\}$ ,  $j \neq k$ , the factor  $|y - x^\top \theta_k^0| = r \leq cs_k$  on  $W_k$ , so the contribution is at most  $cs_k M_x P_0(z = j, h = k)$  in norm, and summing over  $j \neq k$  gives at most  $cs_k M_x \eta$ . Altogether  $\|\mathbb{E}[x(y - x^\top \theta_k^0)\mathbf{1}_{W_k}]\| \leq (1 - \varepsilon)cs_k M_x (\pi_k \eta_k + \eta) \leq 2(1 - \varepsilon)cs_k M_x \eta$ , and dividing by  $\lambda_{\min}(M_k) \geq (1 - \varepsilon)\pi_k \lambda_k$  gives the bound.

*The special case.* If  $\varepsilon = 0$  and  $\eta = 0$  then  $\delta_k = \nu_k = 0$ , so  $s_k = q(\frac{1}{2})\sigma_k$  by (i),  $\kappa = 2\Phi(-cq(\frac{1}{2}))$  by (ii), and the bias bound is zero.  $\square$

Three comments. First, the constant in the limit of the flagged fraction is  $2\Phi(-cs_k/\sigma_k)$  and not  $2\Phi(-c)$ . On the units assigned to surface  $k$  the minimum-over-surfaces residual is the residual from surface  $k$  itself, so the per-group median is a median of own-surface residuals; it is pulled down because the clean units of group  $k$  with the largest errors are nearer to another surface and are assigned there, and pulled up by the units of other groups and by the contamination assigned to surface  $k$ . Theorem 3(i) bounds the net effect by  $\nu_k = \delta_k + \eta_k/2$ , which is of the order of  $\eta + \varepsilon$ . The first effect is intrinsic to any scale computed within an assigned group and vanishes only when the groups do not overlap. Second, the refit is exactly unbiased at  $\theta^0$  when the groups do not overlap and moves by  $O(\eta)$  when they do; the source is the asymmetric truncation of the error on  $\{h = k\}$ , not the contamination, which the step removes entirely under Assumption 2. Third, Algorithm 1 reports the flagged fraction recomputed at the refitted parameter, that is  $\hat{\alpha}_n(\hat{\theta})$  rather than  $\hat{\alpha}_n(\theta)$ ; Theorem 2 concerns the flag set the refit uses. The limit of the reported fraction is  $\alpha(\theta^\infty(\theta))$  whenever  $\alpha$  is continuous at  $\theta^\infty(\theta)$  and the convergence of  $\hat{\alpha}_n$  is uniform near it, which we do not prove.

#### A.4 The group-recovery step

The local peak test guards against a slice of a band of gross outliers, which the likelihood would value like a group. Let  $W = 3cs$ ,  $u = 1/(2W)$ ,  $g_W$  the  $\mathcal{N}(0, s^2)$  density truncated to  $[-W, W]$ , and

for a residual density  $p$  on the window let  $w^*(p)$  maximise  $L_p(w) = \int p \log\{wg_W + (1-w)u\}$  over  $[0, 1]$ .

**Proposition 4.** (i)  $L_p$  is strictly concave, so  $w^*(p)$  is unique, and the EM iteration in  $w$  started at  $\frac{1}{2}$  converges monotonically to it; the same holds for the sample version, whose maximiser converges to  $w^*(p)$  almost surely for independent residuals of density  $p$ .

(ii) If  $|r|$  is stochastically smaller under  $\tilde{p}$  than under  $p$ , then  $w^*(\tilde{p}) \geq w^*(p)$ .

(iii) If  $p = u$ , then  $w^*(p) = 0$ ; if  $p$  is uniform on  $[-V, V]$ ,  $V \leq W$ , then  $w^*$  is nonincreasing in  $V$ .

(iv) If  $p = w_0 g_W + (1 - w_0)u$ , then  $w^*(p) = w_0$ .

The test thus passes a group whose units are at least half of those in the window and fails a slice of a band that fills it. Between the two, solving  $L'_p(w) = 0$  for  $c = 2.5$ , a uniform band of half-width  $V$  has  $w^* \geq \frac{1}{2}$  exactly when  $V \leq 3.44s$ : a band narrower than about seven noise scales passes the test and is left to the scale restriction and the likelihood. The implementation uses the untruncated normal density, which differs from  $g_W$  by the mass  $2\Phi(-3c) \approx 6 \times 10^{-14}$  outside the window; (ii) and (iii) hold for it as stated. The proposition concerns a given residual density: the effect of the RANSAC selection on  $p$  is not analysed, and the fifty EM iterations of the implementation approximate the limit in (i).

*Proposition 3(i).* Write  $f = f_F$  and  $f' = f_{F'}$ , so that  $f' = f - \pi_*/R + (\pi_*/\sigma_*)\varphi((y - x^\top \theta_*)/\sigma_*)$ , and split  $L(F') - L(F) = \mathbb{E}[(\log f' - \log f)\mathbf{1}_G] + \mathbb{E}[(\log f' - \log f)\mathbf{1}_{G^c}]$ .

On  $G$ ,  $y - x^\top \theta_* = \sigma_* e$ , and dropping the lines of  $F$  from  $f'$ ,

$$f' \geq \frac{\pi_*}{\sigma_*} \varphi(e) + \frac{\pi_0 - \pi_*}{R} = \frac{\pi_*}{\sigma_*} \varphi(e)(1 + \tau e^{e^2/2}), \quad \tau = \frac{(\pi_0 - \pi_*)\sigma_* \sqrt{2\pi}}{\pi_* R}.$$

Since  $\mathbb{E} \log \varphi(e) = -\frac{1}{2} \log(2\pi) - \frac{1}{2} \mathbb{E} e^2 = -\frac{1}{2} \log(2\pi e)$ ,

$$\mathbb{E}[\log f' \mid G] \geq \log \frac{\pi_*}{\sigma_*} - \frac{1}{2} \log(2\pi e) + \rho, \quad \rho = \mathbb{E} \log(1 + \tau e^{e^2/2}) \geq 0.$$

For  $v \geq 0$ ,  $\log(1 + v) \leq 2\sqrt{v}$  (the difference vanishes at 0 and has derivative  $v^{-1/2} - (1 + v)^{-1} > 0$ ), and  $\mathbb{E} e^{e^2/4} = (1 - \frac{1}{2})^{-1/2} = \sqrt{2}$ ; hence  $\rho \leq 2\sqrt{\tau} \mathbb{E} e^{e^2/4} = 2\sqrt{2\tau}$ . Next,  $f = (\pi_0/R)\{1 + (R/\pi_0) \sum_k (\pi_k/s_k) \varphi((y - x^\top \theta_k)/s_k)\}$ , so  $\mathbb{E}[\log f \mid G] = \log(\pi_0/R) + \xi$ . By Jensen's inequality,  $\xi \leq \log\{1 + (R/\pi_0) \sum_k \pi_k \mathbb{E}[s_k^{-1} \varphi((y - x^\top \theta_k)/s_k) \mid G]\}$ , and on  $G$ ,  $y - x^\top \theta_k = x^\top (\theta_* - \theta_k) + \sigma_* e$  with  $e$  independent of  $x$ ; integrating over  $e$  first, the convolution of two centred normal densities gives  $\mathbb{E}[s_k^{-1} \varphi((y - x^\top \theta_k)/s_k) \mid x, G] = \varphi_k(x^\top (\theta_* - \theta_k))$ , which is the stated bound. Multiplying by  $P(G) = \pi_*$ , the first expectation is at least  $\pi_*$  times the brace in (3) plus  $\pi_* \rho$ ; the statement drops  $\rho \geq 0$ , and the bound  $\rho \leq 2\sqrt{2\tau}$  shows that the leading term is exact up to  $\xi$  and the two dropped nonnegative terms,  $\rho$  and the density of the lines of  $F$  on  $G$  in  $f'$ .

On  $G^c$ ,  $f' \geq f - \pi_*/R > 0$  because  $f \geq \pi_0/R > \pi_*/R$ , so  $\log f' - \log f \geq -\log\{f/(f - \pi_*/R)\}$  and the second expectation is at least  $-\Lambda$ ,  $\Lambda = \mathbb{E}[\log\{f/(f - \pi_*/R)\}\mathbf{1}_{G^c}]$ . With  $t = (\pi_0/R)/f \in (0, 1]$  and  $b = \pi_*/\pi_0 < 1$ ,  $\log\{f/(f - \pi_*/R)\} = -\log(1 - bt)$ ; the function  $t \mapsto -\log(1 - bt)$  is convex on  $[0, 1]$  and vanishes at 0, so it is at most  $t$  times its value at 1, namely  $t \log\{\pi_0/(\pi_0 - \pi_*)\}$ . Taking expectations over  $G^c$  gives  $\Lambda \leq \nu \log\{\pi_0/(\pi_0 - \pi_*)\}$ , and  $\nu \leq P(G^c)$  since  $t \leq 1$ .  $\square$

*Proposition 3(ii).* On  $S$ , the residuals from the two lines are  $\sigma_S e \mp \delta/2$ , so their joint density is

$$A = \frac{\pi_S}{2\sigma_S} \{\varphi(e - a) + \varphi(e + a)\} = \frac{\pi_S}{\sigma_S} \varphi(e) e^{-a^2/2} \cosh(ae), \quad \text{against} \quad B = \frac{\pi_S}{\sigma_S} \varphi(e)$$

for the single line, and  $\log(B/A) = a^2/2 - \log \cosh(ae)$ . Let  $r \geq 0$  be the density of the other components, the same in both fits. Off  $S$  the two densities coincide, and on  $S$ ,  $\log f_{F''} - \log f_F = \psi$  with  $\psi = \log\{(B+r)/(A+r)\}$ . Since  $t \mapsto \log\{(B+t)/(A+t)\}$  is monotone on  $[0, \infty)$  with limit 0,  $\psi$  lies between  $\min\{\log(B/A), 0\}$  and  $\max\{\log(B/A), 0\}$ . Hence

$$-\pi_S D_-(a) \leq L(F'') - L(F) = \pi_S \mathbb{E}[\psi \mid S] \leq \pi_S D_+(a),$$

where  $D_\pm(a) = \mathbb{E}[(\pm \frac{1}{2}a^2 - \log \cosh(ae))^+]$ . Two inequalities for  $x \in \mathbb{R}$ :  $\log \cosh x \leq x^2/2$ , because both sides vanish at 0 and  $\tanh x \leq x$  for  $x \geq 0$ ; and  $\log \cosh x \geq x^2/2 - x^4/12$ , because the difference vanishes at 0 and its derivative  $\tanh x - x + x^3/3$  is nondecreasing on  $[0, \infty)$ , its own derivative being  $x^2 - \tanh^2 x \geq 0$ . By the first,  $D_-(a) \leq \mathbb{E}[(a^2 e^2/2 - a^2/2)^+] = \frac{1}{2}a^2 \mathbb{E}(e^2 - 1)^+ = \varphi(1)a^2$ , since  $\mathbb{E}(e^2 - 1)^+ = 2 \int_1^\infty (z^2 - 1)\varphi(z) dz = 2\varphi(1)$ . Further  $D_+(a) - D_-(a) = \mathbb{E} \log(B/A) = \mathbb{E} \log\{\varphi(e)/\frac{1}{2}(\varphi(e-a) + \varphi(e+a))\}$  is the Kullback–Leibler divergence stated, hence nonnegative, and by the second inequality it is at most  $\mathbb{E}[a^4 e^4]/12 = a^4/4$ . So  $D_+(a) \leq \varphi(1)a^2 + a^4/4$ , which gives the two bounds. Finally, if  $r$  is replaced by  $tr$ , then  $\psi \rightarrow \log(B/A)$  pointwise as  $t \downarrow 0$ , dominated by  $|\log(B/A)| \leq a^2 e^2/2 + a^2/2$ , which is integrable; by dominated convergence the difference tends to  $\pi_S \mathbb{E} \log(B/A)$ .  $\square$

*Acceptance.* If  $F_1$  arises from  $F_0$  by (i) and then (ii), then  $L(F_1) - L(F_0)$  is at least the right side of (3) minus  $\varphi(1)a^2\pi_S$ , which is positive by assumption. By the strong law, applied to the bounded summands  $\log f_{F_1}(u_i)$  and  $\log f_{F_0}(u_i)$ ,  $\{\ell_n(F_1) - \ell_n(F_0)\}/n \rightarrow L(F_1) - L(F_0) > 0$  almost surely, while  $\frac{d+1}{2} \log n = o(n)$ .  $\square$

*Proposition 4.* On the window,  $g_W$  lies between  $\varphi(3c)/(s\kappa)$  and  $\varphi(0)/(s\kappa)$  with  $\kappa = 1 - 2\Phi(-3c)$ , and  $u > 0$ , so  $\log\{wg_W + (1-w)u\}$  is bounded on  $[-W, W] \times [0, 1]$  and  $L_p$  is finite. Write  $q_w = wg_W + (1-w)u$ .

(i). For fixed  $r$ ,  $w \mapsto \log q_w(r)$  is concave, and strictly so unless  $g_W(r) = u$ , which happens for at most two values of  $r$ , a  $p$ -null set. Hence  $L_p$  is strictly concave and has a unique maximiser. Its derivative  $L'_p(w) = \int p(g_W - u)/q_w$  is continuous and strictly decreasing, so  $w^* = \sup\{w \in [0, 1] : L'_p(w) \geq 0\}$ , the supremum of the empty set being 0. The EM map is  $M(w) = \int p w g_W / q_w$ , and

$$M(w) - w = \int p \frac{w g_W - w q_w}{q_w} = w(1-w)L'_p(w), \quad M'(w) = \int p \frac{g_W u}{q_w^2} > 0.$$

Thus  $M$  is increasing, its fixed points are 0, 1 and the zero of  $L'_p$  in  $(0, 1)$  when there is one, and  $M(w) > w$  exactly when  $L'_p(w) > 0$ , that is when  $w < w^*$ . Start at  $w_0 = \frac{1}{2}$ . If  $w_0 < w^*$ , then  $w_0 < M(w_0) \leq M(w^*) = w^*$  when  $w^* < 1$ , and  $M(w_0) \leq 1$  otherwise; by induction the iterates increase and stay at most  $w^*$ , so they converge to a fixed point in  $(\frac{1}{2}, w^*]$ , which is  $w^*$ . If  $w_0 > w^*$ , they decrease and stay at least  $w^*$ , and converge to a fixed point in  $[w^*, \frac{1}{2})$ , which is  $w^*$  when  $w^* > 0$  and 0 when  $w^* = 0$ . The sample criterion  $L_N(w) = N^{-1} \sum_{i \leq N} \log q_w(r_i)$  has the same properties, with  $p$  replaced by the empirical law, as soon as one  $r_i$  has  $g_W(r_i) \neq u$ , which is almost sure when the  $r_i$  have a density; the implemented iteration is its EM map, so the same argument applies to its maximiser  $\hat{w}_N$ . If the  $r_i$  are independent with density  $p$ , then  $L_N(w) \rightarrow L_p(w)$  almost surely for each  $w$  in a countable dense subset of  $[0, 1]$ , by the strong law for bounded summands; the functions  $w \mapsto \log q_w(r)$  have derivative bounded by  $\max(g_W, u)/\min(g_W, u)$  uniformly in  $r$ , so the  $L_N$  are equi-Lipschitz and the convergence is uniform on  $[0, 1]$ ; uniform convergence to a continuous function with a unique maximiser on a compact set gives  $\hat{w}_N \rightarrow w^*(p)$ .

(ii). For  $t \geq 0$  put  $h_w(t) = (g_W(t) - u)/(wg_W(t) + (1-w)u)$ . As a function of  $g \in (0, \infty)$ ,  $(g - u)/(wg + (1-w)u)$  has derivative  $u/(wg + (1-w)u)^2 > 0$ , and  $g_W(t)$  is decreasing in  $t$ , so  $h_w$  is decreasing and bounded. Since  $L'_p(w) = \mathbb{E}_p[h_w(|r|)]$ , the stochastic order gives  $L'_p(w) \geq L'_p(w)$

for every  $w$ , hence  $\{w : L'_p(w) \geq 0\} \supseteq \{w : L'_p(w) \geq 0\}$  and  $w^*(\tilde{p}) \geq w^*(p)$  by the characterisation in (i).

(iii). If  $p = u$ , then  $L'_p(0) = \int u(g_W - u)/u = \int_{-W}^W g_W - 1 = 0$ , so  $L'_p(w) < 0$  for  $w > 0$  and  $w^* = 0$ . With the untruncated normal density  $g$  in place of  $g_W$  the same computation gives  $L'_p(0) = \kappa - 1 < 0$ , and again  $w^* = 0$ . If  $p$  is uniform on  $[-V, V]$ , then  $|r|$  is stochastically smaller for smaller  $V$ , and (ii) applies; the proof of (ii) uses only that the peak density is decreasing in  $|r|$ , so it holds with  $g$  in place of  $g_W$  as well.

(iv). If  $p = w_0 g_W + (1 - w_0)u = q_{w_0}$ , then  $L'_p(w_0) = \int (g_W - u) = 0$ , so  $w_0$  is the zero of  $L'_p$  and  $w^* = w_0$ .  $\square$

*Numerical values.* The constants quoted in Section 3.3 were obtained by solving  $L'_p(w) = 0$  by quadrature and bisection with  $s = 1$  and  $c = 2.5$ . For the uniform band of half-width  $V$ ,  $w^*$  is 0, 0.08, 0.18, 0.34, 0.71 and 1 at  $V = 7.5, 6, 5, 4, 3$  and  $2.5$ , and equals  $\frac{1}{2}$  at  $V = 3.44$ . For a slice whose density on  $|r| \leq cs$  exceeds that on the rest of the window by a factor  $1 + \beta$ , as the RANSAC selection tends to produce,  $w^*$  reaches  $\frac{1}{2}$  at  $\beta = 5.2$ , that is when the central third of the window holds 76% of its units against 33% for a uniform slice. For a mixture of a Gaussian share  $w_0 \in \{0.3, 0.5, 0.7\}$  and a uniform background the computed maximiser equals  $w_0$  to six decimals, with  $g_W$  and with the untruncated density alike. The bounds of Proposition 3 were checked by quadrature and by Monte Carlo with  $2 \times 10^6$  units: with  $R/\sigma_* = 30$ ,  $\pi_* = 0.1$ ,  $\pi_0 = 0.15$  and two lines seven scales away, the mean gain per unit of  $G$  was 1.6695 against a leading term of 1.5768,  $\rho = 0.0928$  and  $\xi \leq 1.2 \times 10^{-4}$ ; with the lines three scales away the gain fell to 0.90 and the Jensen bound on  $\xi$  was 1.85 against a true value of 1.47. For the un-splitting,  $D_+ - D_-$  and  $D_-$  were within their bounds at  $a \in \{0.1, 0.25, 0.5, 1\}$ , and with a noise floor present the mean change per unit of  $S$  at  $a = 0.5$  was 0.016, inside  $[-0.049, 0.061]$ , against 0.012 for the divergence.

## B Implementation

**Exact subsample fits.** Label vectors are enumerated depth first in restricted-growth form: unit  $j$  may receive any label already used by units  $1, \dots, j-1$  or the smallest unused one. This lists each partition into at most  $K$  groups exactly once and removes the  $K!$  relabellings. Each group keeps its sufficient statistics  $(n_k, X_k^\top X_k, X_k^\top y_k, y_k^\top y_k)$ , updated in  $O(d^2)$  when a unit enters or leaves, and its residual sum of squares, recomputed by a Cholesky solve of size  $d$ . For rank-deficient groups the minimum-norm solution is used; its residual sum of squares is the minimum over all coefficient vectors, so the group criterion is exact in every case. A group's residual sum of squares cannot decrease when a unit is added, so the sum over groups at a partial labelling is a lower bound for every completion, and a branch is abandoned when the bound reaches the best value found. The cost of the search is reported in Section 2 (Table 1).

The code was checked against plain enumeration of all  $K^m$  label vectors, with residual sums of squares computed independently by a least-squares routine, on 200 instances with  $(K, d, m) \in \{(2, 1, 10), (2, 2, 10), (3, 2, 8), (2, 3, 9)\}$ , a third of them with tied covariates.

**Concentration steps.** Every unit is assigned to its nearest surface, the  $h$  units with the smallest squared residuals are kept, and each group is refitted on its kept units; this is repeated until the trimmed sum of squares stops decreasing, at most 200 times.

**Software.** The procedure is written in Python with `numpy`, `scipy` and `numba`, and the covariate screen uses the minimum covariance determinant of `scikit-learn` when there is more than one

covariate. The code, the scripts that produce every table and figure, and the raw results are at <https://github.com/sorujov/eesp>.